

Table of Contents

ZAREWA ALUMINIUM AND PLASTICS LTD

COMPLETE SYSTEM OF OPERATIONS — MASTER DOCUMENT

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ZAREWA ALUMINIUM AND PLASTICS LTD

SYSTEM OF OPERATIONS — COMPLETE PROCEDURES, POLICIES & STANDARDS

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PURPOSE OF THIS MANUAL

This System of Operations (SOP) manual is the authoritative reference for how Zarewa Aluminium and Plastics Ltd conducts business through its integrated Enterprise Resource Planning (ERP) system. Every procedure described herein maps directly to screens, permissions, workflows, and business rules implemented in the live software. Staff must follow these procedures to ensure:

1. **Financial integrity** — every naira in and out is recorded, cleared, reconciled, and auditable.
2. **Operational control** — stock, coils, production, and deliveries are tracked from purchase to customer handover.
3. **Segregation of duties** — no single person can request, approve, and pay the same transaction.
4. **Regulatory and board compliance** — payroll, tax, pension, discipline, and exit procedures meet company policy.
5. **Branch accountability** — Kaduna HQ, Yola Factory, and Maiduguri Factory each maintain accurate records within a unified system.

This manual is not a generic industry template. It describes **your actual system** — the quotations you create in /sales, the receipts you confirm in /accounts, the production jobs you complete in /operations, and the approvals that flow through /manager and /exec.

COMPANY IDENTITY

Legal name: Zarewa Aluminium and Plastics Ltd

Trading name: Zarewa Industries

Industry: Manufacturing and distribution of roofing sheets, stone-coated tiles, and building accessories

Headquarters: Kaduna, Nigeria

Active branches (system-managed):

Branch

Kaduna

Yola

Maiduguri

Products and Services

Zarewa manufactures and sells:

- **Long-span and step-tile aluminium/aluzinc roofing sheets** — multiple gauges (0.18 mm to 0.55 mm standard) and colour profiles
- **Stone-coated steel roofing tiles** — stone-flatsheet format with tracked sheet-to-m² conversion
- **Roofing accessories** — ridges, flashings, gutters, fixings, and related consumables
- **Outside corrugation** — sub-contracted to approved third parties where required
- **Transport and installation services** — quoted separately on customer quotations where applicable

All products are measured and tracked in **metres (m)** for sheet production, **kilograms (kg)** for coil raw material, and **NGN (₦)** for all financial values.

DOCUMENT NUMBERING STANDARD

Every business document in the ERP carries a human-readable ID following the format:

PREFIX-BRANCHCODE-YY-NNNN

Where YY is the two-digit calendar year and NNNN is a sequential number resetting each year per branch. Legacy IDs without branch codes are still recognised by the system to prevent collisions during migration.

Prefix

QT

CL

PJ

DLV

LE

RCP

PO

IT

MINT

MREQ

CREV

CREQ

BD

BDA

TM

SM

EXP

WI

OBL
MACH
MXPL
MXWO
MXEV

Office filing references on approved work items may additionally receive:
ZR/{branch}/{domain}/{year}/{seq}.

MANUAL STRUCTURE — DEPARTMENT SOPs

This master manual is organised into standalone department SOPs. Each SOP is self-contained but cross-references related procedures.

File	Title	Primary Users
01-COMPANY- GOVERNANCE- AND-SYSTEM.md	Company governance, ERP architecture, security, cross-cutting rules	All staff, IT admin
SOP-01-SALES- OFFICE.md	Sales office — customers, quotations, cutting lists, CRM	Sales officers, branch managers
SOP-02- CASHIER- DESK.md	Cashier desk — receipts, advances, bank deposits, payouts	Cashiers
SOP-03- ACCOUNTING- DESK.md	Accounting desk — GL, reconciliation, month-end, AP phases	Accountants, finance manager
SOP-04- OPERATIONS- STORE.md	Operations & store — GRN, coil register, stock, deliveries	Operations officers, storekeepers
SOP-05- PRODUCTION.m d	Production floor — jobs, conversion QC, WIP, revenue recognition	Production managers, QCO
SOP-06- PROCUREMENT.	Procurement — POs, suppliers,	MD, procurement officer, operations

md	transport, in-transit	
SOP-07-HUMAN-RESOURCES.md	HR — leave, loans, payroll, discipline, exit	HR admin, GM HR, all staff
SOP-08-EXECUTIVE-OFFICE.md	Executive office — MD cockpit, approvals, governance	MD, CEO, Chairman
SOP-09-MAINTENANCE.md	Maintenance — machines, work orders, asset custody	Maintenance manager, technicians
SOP-10-OFFICE-ADMINISTRATION.md	Office administration — memos, work items, notices, forum	All staff with office.use
APPENDIX-A-GLOSSARY-AND-REFERENCE.md	Glossary, GL codes, permission bundles, report index	Reference
ANNEX-B-SCENARIO-WALKTHROUGH S.md	10 end-to-end worked business scenarios	Training
ANNEX-C-IT-OPERATIONS.md	IT admin, security, DR, deployment	IT Admin
ANNEX-D-COMPLIANCE-AND-AUDIT.md	Internal controls, audit, fraud response	Audit, Finance, MD
ANNEX-E-EXTENDED-PROCEDURES.md	Troubleshooting, escalation matrix, training	All departments
ANNEX-F-ACCOUNTING-POLICIES.md	AP1c/AP2/AP3, GL, period close	Finance
ANNEX-G-HR-POLICIES.md	HR policy manual, discipline, exit	HR, all staff
ANNEX-H-INVENTORY-PRODUCTION-STANDARDS.md	Coil science, QC, stock register	Operations, Production

Combined master document: Run node docs/SOP/build-master-sop.mjs to generate ZAREWA_COMPLETE_SOP_v3.md (~30,000+ words).

QUICK-START BY ROLE

After signing in, the system routes each user to their role home screen:

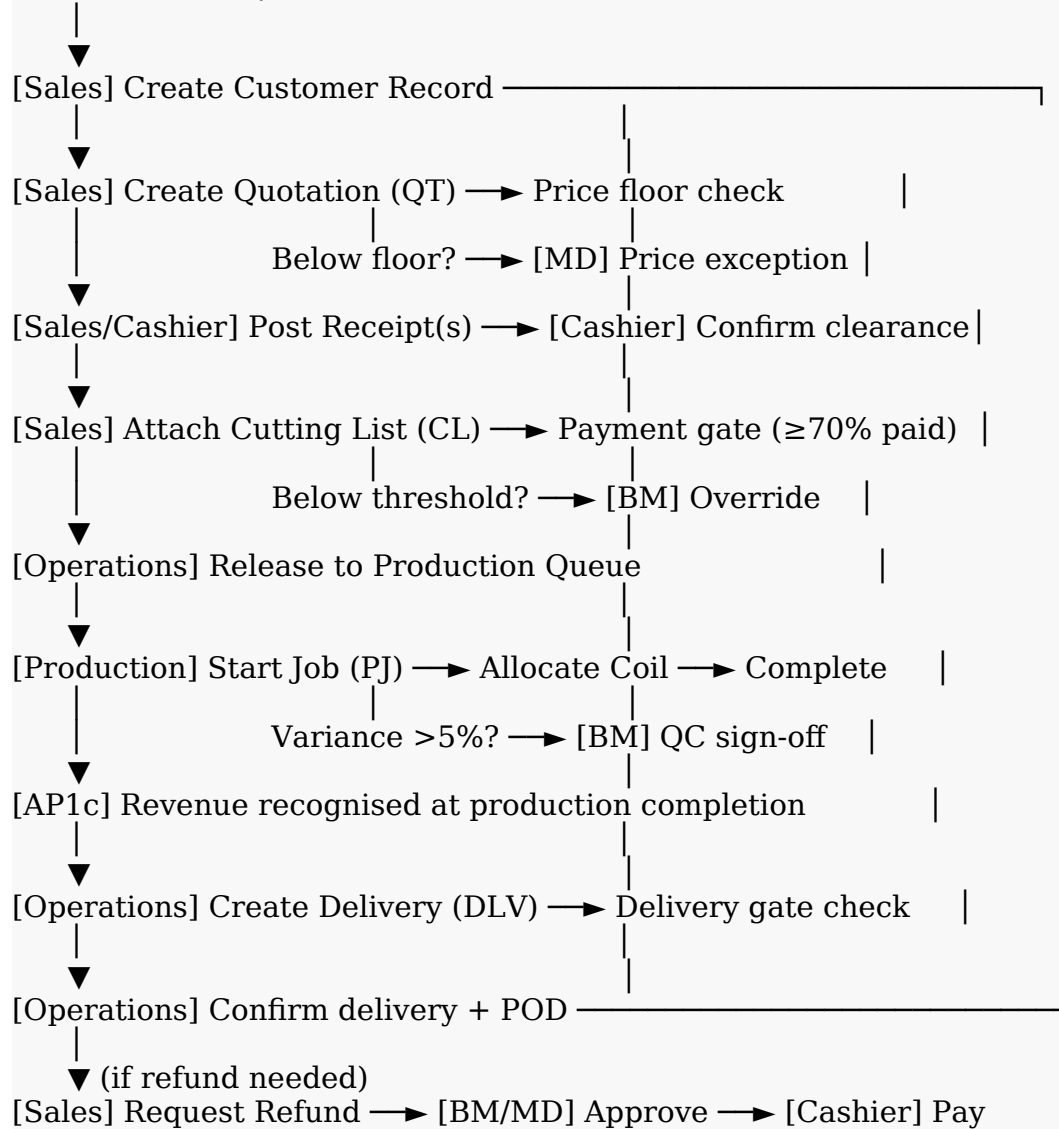
Role	System Key	Default Home	Primary SOP
Managing Director	md	/exec (Command Centre)	SOP-08
CEO	ceo	/exec (read-only)	SOP-08
Chairman	chairman	/exec + Executive HR	SOP-08, SOP-07
Branch Manager	sales_manager	/manager	SOP-01, SOP-04
Sales Officer	sales_staff	/sales	SOP-01
Cashier	cashier	/accounts	SOP-02
Accountant / Head of Accounts	finance_manager	/accounting	SOP-03
Operations Officer / Storekeeper	operations_officer	/operations	SOP-04, SOP-05
HR Admin	hr_admin	/hr/dashboard	SOP-07
GM HR Admin	gmhr	/hr/dashboard	SOP-07
Administrator	admin	/ or /settings	All SOPs + 01
HR Portal Only	hr_portal_only	/my-profile	SOP-07 (self-service)
Viewer	viewer	/ (read-only)	01 (overview only)

First login: All new users must change their password on first sign-in. A role-specific training guide appears automatically. Use **Zare** (life-ring icon, bottom-right) on any screen for step-by-step coaching — Zare explains procedures; you perform every action yourself.

CORE BUSINESS LIFECYCLE (END-TO-END)

The following diagram shows how a typical customer order flows through the entire organisation:

CUSTOMER INQUIRY



SEGREGATION OF DUTIES — MASTER REFERENCE

Process	Create / Request	Approve / Confirm	Pay / Execute
Customer receipt	Sales / Cashier (receipts.post)	Cashier confirms clearance (cashier.receipts.confirm)	Treasury auto-posted
Customer	Sales (refunds.request)	BM / MD / Finance	Cashier / Finance

refund		(refunds.approve)	(finance.pay)
Payment request (expense)	Any authorised staff	BM (≤₦200k) or MD (>₦200k)	Finance pays approved
Below-floor price → production	Sales saves quotation	MD (md.price_exception.approve)	—
Production gate override	—	BM (approve_production with note)	—
Material incident	Operations (material_incidents.create)	BM (material_incidents.approve)	Auto-post on approve
Purchase order	Procurement / MD	MD / BM per policy	—
Supplier payment	Finance	Finance (finance.approve)	Finance (finance.pay)
Payroll	HR Admin prepares	GM HR + MD sign-off	Finance export/pay
Leave / loan request	Employee self-service	HR → BM endorse → GM HR	—
Inter-branch loan	Finance proposes	MD (inter_branch_loan.md_approve)	Treasury
Credit exception	Sales requests	BM or MD per amount	—
Edit on locked record	Requester	BM/MD issues 6-digit code	Single save consumed
Stock register lock	Store confirms	BM → MD → Admin locks	—

Critical rules enforced in code: - Refund requester cannot approve their own refund - Refund approver cannot pay the same refund (when ENFORCE_DUAL_CONTROL_PAYMENTS=1) - Refunds above ₦1,000,000 require MD/CEO approval - Cashier cannot approve refunds — only request and pay after approval - Payroll cannot be locked without MD approval recorded

DAILY, WEEKLY, AND MONTHLY CADENCE

Daily (Every Working Day)

Time	Who	Action	System Location
Opening	Cashier	Review pending receipt queue	/accounts?tab=desk
Opening	Sales	Check quotation follow-up alerts	/sales → Quotations
Opening	Operations	Review production queue and coil availability	/operations
Opening	Branch Manager	Clear Manager dashboard inbox	/manager
Throughout	All	Process office work items within SLA	/ (Workspace)
Closing	Cashier	Reconcile physical cash to treasury balance	/accounts
Closing	Cashier	Confirm all day's receipts	/accounts?tab=receipts

Weekly

Who	Action	System Location
Branch Manager	Review aged pending approvals	/manager
Finance Manager	Review bank reconciliation unmatched lines	/accounting → Reconciliation
Finance Manager	Review payment exception queue	/reports
MD	Review Command Centre alerts and Decide tab	/exec
HR Admin	Process leave/loan request queue	/hr/time-absence
Operations	Review low-stock alerts	/operations → Overview

Monthly (Month-End Close)

Step	Owner	SOP Reference
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1. Complete all GRNs for received goods	Operations	SOP-04, SOP-06
2. Complete all production jobs and deliveries	Production / Operations	SOP-05
3. Stock register 4-stage sign-off (all branches)	Store → BM → MD → Lock	SOP-04 \$4
4. Bank reconciliation (all treasury accounts)	Finance Manager	SOP-03 \$5
5. Payables reconciliation (PO vs GRN vs paid)	Finance Manager	SOP-03 \$6
6. Receivables review and write-offs	Finance + MD	SOP-03 \$7
7. Depreciation run	Finance Manager	SOP-03 \$8
8. Payroll processing (if pay period)	HR → GM HR → MD → Finance	SOP-07 \$4
9. GL trial balance verification	Finance Manager	SOP-03 \$9
10. Period lock	Finance Manager	SOP-03 \$10
11. Management reports pack	Finance / MD	SOP-08 \$3

HOW TO USE THIS MANUAL WITH THE ERP

1. **Find your role** in the Quick-Start table above and read your primary SOP first.
2. **Use Zare** for on-screen coaching: open any page, click the life-ring icon, choose “Tour this page.”
3. **Cross-reference document IDs** — when this manual mentions a QT, CL, or PO, look up that ID in the system search (/ → Search).
4. **Report gaps** — if a procedure in the system behaves differently from this manual, report to IT immediately; the manual will be updated after code verification.

5. **Training** — new staff complete role training guide on first login; supervisors verify competency using the checklists at the end of each department SOP.
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RELATED TECHNICAL DOCUMENTATION

Document	Location	Audience
Operations Manual	Zarewa-backend-main/ docs/ OPERATIONS_MANUAL. md	IT, power users
Access Control	Zarewa-backend-main/ docs/ ACCESS_CONTROL.md	IT, admin
RBAC Matrix	Zarewa-backend-main/ docs/RBAC_MATRIX.md	IT, admin
Refund Operations	Zarewa-backend-main/ docs/ REFUND_OPERATIONS. md	Finance, sales, BM
Accounting Policy AP1c	Zarewa-backend-main/ docs/ ACCOUNTING_POLICY_ AP1C.md	Finance
Accounting Policy AP3	Zarewa-backend-main/ docs/ ACCOUNTING_POLICY_ AP3_COSTING.md	Finance, production
Payment Posting SOP	Zarewa-backend-main/ docs/ PAYMENT_POSTING_SO P.md	Cashier, finance
Material Exceptions SOP	Zarewa-backend-main/ docs/ MATERIAL_EXCEPTION S_SOP.md	Operations, BM
HR Policy — Leave	Zarewa-backend-main/ docs/HR/HR-POLICY- LEAVE.md	HR
HR Policy — Payroll	Zarewa-backend-main/ docs/HR/HR-POLICY- PAYROLL.md	HR, finance
Deployment Guide	Zarewa-backend-main/ docs/DEPLOYMENT.md	IT admin

End of Master Index. Proceed to individual department SOPs for detailed procedures.

PART 1: COMPANY GOVERNANCE, ERP ARCHITECTURE & CROSS-CUTTING RULES

Zarewa Aluminium and Plastics Ltd — System of Operations v3.0

Applies to: All staff, IT administrators, external auditors

System modules: Platform-wide

1. INTRODUCTION

1.1 Scope

This section establishes the governance framework within which all Zarewa business operations are conducted through the integrated ERP system. It covers company identity, organisational structure, technology architecture, authentication and security controls, branch data isolation, document standards, approval thresholds, audit requirements, and the cross-cutting business rules that apply regardless of department.

Every staff member — from the sales officer creating a quotation in Kaduna to the production manager completing a job in Yola — operates within the boundaries defined here. Department-specific procedures in SOP-01 through SOP-10 build upon this foundation.

1.2 Authority

The Managing Director (MD) holds ultimate operational authority. The Board of Directors, through the Chairman and CEO, exercises strategic oversight via the Executive Command Centre. Day-to-day system administration is delegated to the System Administrator (admin role). Financial period control rests with the Accountant / Head of Accounts (finance_manager role).

1.3 Compliance Obligation

All staff are required to:

1. Use the ERP for every business transaction — no parallel paper-only processes for items the system tracks.
 2. Operate only within their assigned role permissions — attempting to access unauthorised modules is logged.
 3. Complete mandatory training before independent transaction entry.
 4. Report system errors, discrepancies, or suspected misuse immediately to their line manager and IT.
-

2. ORGANISATIONAL STRUCTURE

2.1 Branch Network

Zarewa operates three active sites, each configured as a branch in the ERP:

Kaduna (BR-KD) — Headquarters - Primary sales office and commercial hub - Default workspace branch for new users unless otherwise assigned - Hosts executive functions, procurement HQ, and central pricing administration - Cutting list minimum paid fraction: 70% (configurable in branches table)

Yola Factory (BR-YOL) — Manufacturing - Primary production facility for coil rolling and corrugation - Receives cutting lists from Kaduna sales; dispatches finished goods - Maintains independent coil register, WIP balances, and stock register - GRN posts against procurement POs received at Yola

Maiduguri Factory (BR-MAI) — Manufacturing - Secondary production and distribution point - Same operational modules as Yola - Inter-branch material requests may transfer stock between factories

2.2 Functional Desks

The ERP organises work by functional desk, each mapped to system modules and role permissions:

Desk	System Route	Primary Roles
Sales Office	/sales, /customers	sales_staff, sales_manager
Cashier Desk	/accounts (Desk, Receipts tabs)	cashier
Accounting Desk	/accounting	finance_manager
Procurement	/procurement	md, operations_officer
Operations / Store	/operations	operations_officer
Production Floor	/operations → Production line	operations_officer, production manager
Branch Management	/manager	sales_manager
Executive Office	/exec	md, ceo, chairman
HR Administration	/hr/*	hr_admin, gmhr
Team HR	/team-hr/*	sales_manager (line managers)
Executive HR	/executive-hr/*	md, ceo, chairman
Self-Service	/my-profile/*	all staff

HR		
Office Administration	/ (Workspace)	all with office.use
System Administration	/settings/*	admin, md (limited)

2.3 Reporting Lines

Operational reporting follows the designation hierarchy encoded in HR:

- **Managing Director** (desig_md) — ultimate approver for high-value transactions, payroll, inter-branch loans, price exceptions
- **Executive Directors** (Operations, Finance, Commercial) — functional oversight within their domains
- **General Manager** (desig_gm) — operational coordination
- **GM HR** (desig_gmhr) — final HR approvals for leave, loans, transfers, exit clearance
- **Branch Manager** (desig_bm, desig_actbm) — branch-level sales, production, refund, and expense approvals
- **Department heads** (Production Manager, Head of Procurement, HR Representative, etc.) — domain-specific supervision

3. ERP SYSTEM ARCHITECTURE

3.1 Technology Stack

Layer	Technology	Purpose
Frontend	React 19 + Vite 8 + Tailwind CSS 4	Single-page application; role-gated routing
Backend API	Node.js + Express 5	REST API with server-side session management
Database	MySQL (production) / SQLite (development)	124+ tables; synchronous write transactions
Authentication	Session cookies + CSRF tokens	bcrypt password hashing; sliding timeout
Shared logic	shared/ directory	Cross-cutting business rules used by API and UI
AI Assistant	Zare (OpenAI/Ollama optional)	In-app coaching, memo polish, operational Q&A
Testing	Vitest (unit) + Playwright (e2e)	Automated workflow verification

3.2 Deployment Models

Combined deployment (recommended for single-server): - API serves the built React SPA from the same origin - Environment variable: ZAREWA_STATIC_DIR points to frontend dist/ - Simplifies cookie and CSRF handling

Split deployment (CDN + API server): - Frontend on static host; API on VM - Requires VITE_API_BASE, CORS_ORIGIN, and careful cookie SameSite configuration - See docs/SPLIT_DEPLOYMENT_AND_MIGRATION.md

Development stack: - npm run dev:stack — API on port 8787, Vite dev server on 5173 with proxy

3.3 Bootstrap Data Model

On login, the SPA loads GET /api/bootstrap — a filtered workspace snapshot containing branches, permissions, master data, and entity lists the user may see. Additional lazy snapshots load on demand:

- GET /api/workspace/sales-snapshot
- GET /api/workspace/operations-snapshot
- GET /api/workspace/finance-snapshot
- GET /api/workspace/procurement-snapshot

Security note: Bootstrap filtering is a performance convenience, not a security boundary. Every API endpoint enforces requirePermission() independently. Calling an API directly without permission returns 403 even if bootstrap omitted the data.

3.4 Data Scope Rules

Data Type	Scope	Notes
Customers, quotations, receipts, coils, production	Branch-scoped	Filtered by branch_id
Suppliers, transport agents	Company-wide	Shared across all branches
Users, roles, org policy	Company-wide	Admin-managed
GL, treasury (with permission)	Branch or org-wide	hq.view_all_branches for rollup
HR staff profiles	Branch-assigned with org-wide HR access	HR admin sees all
Office threads, work items	Branch or company scope	Depends on record type

4. AUTHENTICATION AND SESSION SECURITY

4.1 Sign-In Procedure

1. Navigate to the Zarewa URL provided by IT (e.g., <https://office.zarewa.com> or local dev URL).
2. Enter **username** and **password**.
3. Optional: use **Google sign-in** if Firebase authentication is configured.
4. On success, the system routes to the role default home (see Master Index Quick-Start table).
5. If `must_change_password` flag is set, the user is forced to set a new password before proceeding.
6. Role training guide modal appears on first successful login after password change.

4.2 Password Policy

Control	Requirement
Minimum length	8 characters
Complexity	At least one uppercase, one lowercase, one digit, one special character
First login	Must change temporary password set by admin
Reset	Admin distributes reset code; token TTL 60 minutes
Storage	bcrypt hashed; plaintext never stored or transmitted in logs

4.3 Session Controls

Control	Default	Environment Variable
Session timeout	120 minutes sliding	SESSION_TIMEOUT_MINUTES (range 5-480)
Failed login lock	5 consecutive failures	FAILED_LOGIN_LOCK_THRESHOLD
Lock duration	30 minutes	ACCOUNT_LOCK_MINUTES
CSRF protection	Required on all mutating requests	Automatic
Activity tracking	POST /api/session/activity extends session	Automatic on user interaction

Account lock recovery: Only Administrator or MD can unlock locked accounts via Settings → Team & access. Branch Managers cannot reset passwords or unlock accounts.

4.4 Workspace Branch Selection

1. Confirm the branch shown in the top workspace bar matches your physical location.

2. Admin, MD, CEO, and Chairman may switch branches or enable “view all branches” for org-wide rollup.
3. All writes (quotations, receipts, GRNs, production jobs) post to the **current workspace branch**.
4. Finance with `finance.cross_branch_post` may post receipts to customers in other branches under controlled rules.

Critical rule: Before posting any financial transaction, verbally confirm branch with a colleague if uncertain. Cross-branch mis-posting requires reversal and re-posting with audit trail.

5. ROLE-BASED ACCESS CONTROL (RBAC)

5.1 Permission Model

The system uses granular string permissions (100+ keys) assigned to roles in `server/auth.js` → `ROLE_DEFINITIONS`. Individual users may receive custom overrides in `app_users.permissions_json`, audited via `org_policy_audit`.

Permission check pattern: `requirePermission('permission.key')` on API routes; `ModuleRouteGuard` and desk-specific guards on frontend routes.

5.2 Complete Role Catalogue

Role Key	Display Title	Permission Summary
admin	System Administrator	* (all permissions)
md	Managing Director	Org-wide strategic control, all approvals, pricing, treasury reserve, executive HR
ceo	Chief Executive Officer	<code>exec.dashboard.view</code> + <code>dashboard.view</code> only — read-only executive
chairman	Chairman (Board)	CEO permissions + <code>hr.chairman.manage</code> for family/scholarship accounts
finance_manager	Accountant / Head of Accounts	Accounting desk, GL, reconciliation, treasury oversight, cross-branch posting
cashier	Cashier	Cashier desk only: receipts confirm, treasury payouts, refund pay — no approve
sales_manager	Branch Manager	Sales, operations, production, refund approve, material incident approve, team HR

sales_staff	Sales Officer	Customers, quotations, receipts post, refund request
operations_officer	Operations Officer / Storekeeper	Inventory, production, deliveries, GRN, material incident create
hr_admin	HR Administrator	Full HR: directory, payroll prep, attendance, discipline, exit
gmhr	GM, Human Resources	Org-wide HR final approvals: leave, loans, transfers, exit clearance
hr_portal_only	HR Portal (Self-Service Only)	Self-service only — mining division / scholarship staff
viewer	Read-Only Viewer	dashboard.view only

Legacy aliases: storekeeper and store_keeper normalise to operations_officer.

5.3 Finance Desk Separation (Phase B)

A core internal control separates Cashier and Accounting desks:

Desk	Route	Required Permission	Capabilities
Cashier Desk	/accounts (cashier tabs)	cashier.desk.view, cashier.receipts.confirm	Receive payments, confirm receipts, process payouts, transfers
Accounting Desk	/accounting	accounting.desk.view	GL journals, trial balance, reconciliation, bank deposits, fixed assets, period close

Enforcement: FinanceDeskRouteGuard in frontend; GL API endpoints reject cashier role; legacy tab RBAC in /accounts.

Rationale: One person must not both receive cash and reconcile the bank account. This satisfies basic segregation of duties for cash handling.

5.4 Module Visibility Matrix

Module / Route	Accessible Roles
/sales	All except hr_portal_only, cashier
/customers	sales_manager, sales_staff, finance_manager, md, admin
/procurement	operations_officer, sales_manager, finance_manager, md, admin
/operations	operations_officer, sales_manager, md, admin
/accounts	finance_manager, cashier, md, admin (BM redirected to /manager)

/accounting	finance_manager, md, admin
/manager	sales_manager, admin
/exec	md, ceo, chairman, admin
/hr/*	hr_admin, gmhr
/team-hr/*	sales_manager
/executive-hr/*	md, ceo, chairman
/my-profile/*	all staff
/reports	finance_manager, sales_manager, md, admin, ceo, chairman
/settings	admin, md (limited)
/edit-approvals	sales_manager, md, admin

6. CROSS-CUTTING BUSINESS RULES

6.1 Approval Thresholds

Configurable under Settings → Governance → Office approval thresholds and org_policy_kv:

Threshold

Expense BM approval limit
Refund MD approval threshold
Credit BM limit
Credit MD required above
High-value receipt double-entry
Others expense finance review
AP3 unclassified alert
Others branch coaching

6.2 Payment and Production Gates

Cutting List Payment Gate: - Default: customer must have paid ≥70% of quotation value before cutting list release - Configurable per branch: cutting_list_min_paid_fraction in branches table - Branch Manager may override with documented justification (approve_production on /manager) - Override logged with approver, note, and paid fraction at time of override

Delivery Payment Gate: - Environment: DELIVERY_PAYMENT_GATE = off | warn | enforce - enforce: delivery blocked until balance cleared or active credit exception exists - ALLOW_MD_DELIVERY_OVERRIDE=true: MD may bypass per delivery - DELIVERY_PAYMENT_GATE_STRICT_FINANCE: finance must confirm before dispatch

Below-Floor Price Gate: - Quotation saves below price list floor are allowed with warning
- Cutting list and production blocked until MD records price exception approval - API:
PATCH /api/quotations/:id/md-price-exception-approve

6.3 Accounting Period Control

- All financial writes enforce assertPeriodOpen() for the posting date's period (YYYY-MM)
- Posting to a closed period returns a hard error
- Period locks managed by finance_manager with period.manage permission
- Back-dated corrections require period to be open or admin override

6.4 Edit Approval System

Changes to confirmed/settled documents require Edit Approval:

1. User attempts edit on locked quotation, PO, customer, or delivery record.
2. System prompts edit approval request.
3. Designated approver (BM or MD) opens /edit-approvals.
4. Approver issues **6-digit one-time code** to requester.
5. Requester enters code → single save is consumed.
6. All actions logged in approval_actions and audit_log.

6.5 Audit Trail Requirements

Every significant mutation writes to audit_log with: - Actor user ID and display name - Timestamp (ISO format) - Action type (e.g., refund.create, receipt.bank_confirmation, material_incident.approve) - Target entity type and ID - Branch ID - Optional JSON payload with before/after values

HR-specific actions additionally log to hr_audit_events. Custom permission changes log to customPermissionAudit.

Audit records are immutable — they cannot be deleted or modified by any user including admin.

6.6 Rate Limiting

Authenticated ledger money POSTs (receipt, advance, refund-advance) are rate-limited per user: - Default: 45 requests per minute - Configure: ZAREWA_LEDGER_POST_MAX, ZAREWA_LEDGER_POST_WINDOW_MS

Login and forgot-password endpoints have separate rate limits to prevent brute force.

6.7 Filing Standards

On approval of payment/refund work items, the system may issue filing reference: ZR/{branch}/{domain}/{year}/{seq} stored in work_item_filing.

Filing completeness (filingCompleteness.js) is checked before work items can be formally closed. Documents without filing numbers cannot be archived.

Print snapshots captured at lock/closure stored in work_item_print_snapshots — protects against retrospective document alteration.

7. ORGANISATIONAL POLICY SETTINGS

Stored in org_policy_kv with full audit trail in org_policy_audit:

Policy Key

CREDIT_BRANCH_MANAGER_LIMIT_NGN
CREDIT_MD_REQUIRED_ABOVE_NGN
DELIVERY_PAYMENT_GATE
ALLOW_MD_DELIVERY_OVERRIDE
DELIVERY_PAYMENT_GATE_STRICT_FINANCE
SESSION_TIMEOUT_MINUTES
FAILED_LOGIN_LOCK_THRESHOLD
ACCOUNT_LOCK_MINUTES
OTHERS_MIN_JUSTIFICATION_LEN
OTHERS_FINANCE_REVIEW_THRESHOLD_NGN
AP3_UNCLASSIFIED_ALERT_THRESHOLD_NGN
OTHERS_BRANCH_COACH_THRESHOLD_PCT
maternityLeaveDays

Changes to org policy require MD or admin authority and are permanently audited.

8. OFFLINE AND DEGRADED MODE

If the API becomes unavailable:

1. UI displays “System offline” banner with last synced workspace data.
2. **No new transactions can be saved** — all write operations queue-fail.
3. User should click “Try reconnect” or refresh after IT confirms server health.
4. **Do not** record transactions on paper intending to “catch up later” without supervisor approval — risk of duplicate posting.
5. IT monitors /api/health, /api/readyz, /api/livez endpoints.

Health endpoint exposes DELIVERY_PAYMENT_GATE mode and other feature flags for operational awareness.

9. IT ADMINISTRATION RESPONSIBILITIES

9.1 User Lifecycle

Action	Procedure	System Location
Create user	Set username, role, branch, temporary password	Settings → Team & access
Reset password	Generate reset code; user changes on next login	Settings → Team & access
Unlock account	Clear lock after failed login threshold	Settings → Team & access
Custom permissions	Grant/revoke individual permission overrides	Settings → Team & access (audited)
Deactivate user	Only after HR exit clearance complete (all 5 stages)	Settings → Team & access

9.2 Database and Migrations

- Migrations run automatically on API startup via server/migrate.js
- Schema defined in server/schemaSql.js with incremental migrations
- Production cutover: set ZAREWA_EMPTY_SEED=1 for clean numbering from 0001
- Backup procedures: see docs/DEPLOYMENT.md \$ Backup and recovery

9.3 Regular Maintenance Schedule

Frequency	Task	Owner
Daily	Confirm server uptime; review failed login attempts in audit log	IT Admin
Weekly	Verify database backup integrity; review session activity	IT Admin
Monthly	Database integrity check; review custom permission changes; verify stock register progress	IT Admin + Finance
Quarterly	Rotate admin credentials; audit	IT Admin + MD

	user access list; test disaster recovery	
Per release	Run npm run test and npm run test:e2e; verify migration success	IT Admin

9.4 Integration API Keys

External systems may read trial balance and journal register via Bearer token: - GET /api/integration/v1/trial-balance - GET /api/integration/v1/journals - Keys managed in Settings; read-only access only

10. ZARE AI ASSISTANT

10.1 Purpose

Zare is the built-in AI assistant providing: - Natural language Q&A on ERP workflows - “Tour this page” step-by-step coaching - Memo drafting and polish (/api/office/ai/polish-memo) - Transaction-specific contextual help - Workflow coaching for blocked actions - Business analysis insights for MD and senior staff

10.2 Usage Rules

1. Zare explains procedures — **you perform every action yourself.**
2. Zare does not post transactions, approve items, or override gates on your behalf.
3. Queries are logged in help_query_log; knowledge gaps in help_knowledge_gaps for improvement.
4. User memory (help_user_memory) and branch memory (help_branch_memory) personalise responses.
5. RAG knowledge base in help_rag_chunks weighted by help_article_weights.

10.3 When to Use Zare vs This Manual

Situation	Use
“How do I post a receipt on this screen?”	Zare — Tour this page
“What is the approval chain for refunds above ₦1M?”	This manual § Segregation table
“Why is my cutting list blocked?”	Zare (reads live quotation state)
Month-end close checklist	This manual + SOP-03
Policy interpretation dispute	Line manager + this manual

11. GLOSSARY (KEY TERMS)

Term	Definition
AP1c	Accounting Phase 1c — revenue recognised at production completion with auto GL
AP2	Accounting Phase 2 — received-basis AP; inventory at landed cost
AP3	Accounting Phase 3 — branch P&L, material costing, payroll labour per job
Bootstrap	Initial workspace data snapshot loaded on login
CL	Cutting List — production work order from sales
COGS	Cost of Goods Sold (GL 5010–5050)
Credit Exception	Formal approval for customer credit terms
CSRF	Cross-Site Request Forgery token required on writes
Delivery Gate	Payment check before goods dispatch
GRN	Goods Received Note — store receipt against PO
OBL	Staff Obligation Account — loans, credits, recoveries
POD	Proof of Delivery — customer sign-off on delivery
RBAC	Role-Based Access Control
WIP	Work-in-Progress — goods in production
Zare	Zarewa AI assistant

Full glossary: [APPENDIX-A-GLOSSARY-AND-REFERENCE.md](#)

End of Part 1. Proceed to department SOPs for operational procedures.

SOP-01: SALES OFFICE

Zarewa Aluminium and Plastics Ltd — System of Operations v3.0

Department: Sales Office

System modules: /sales, /customers, /customers/:id, /pricing-policy (read)

Primary roles: sales_staff, sales_manager

Supporting roles: md (price exceptions), cashier (receipt posting)

1. PURPOSE AND SCOPE

1.1 Purpose

The Sales Office is the commercial front door of Zarewa. Every customer relationship, quotation, cutting list, and refund request originates here. This SOP defines how sales officers and branch managers use the ERP to manage the complete sales cycle from first customer contact through quotation, payment tracking, production handoff, and refund initiation.

1.2 Scope

This SOP covers: - Customer registration and CRM - Quotation creation, editing, and lifecycle management - Pricing, floor price exceptions, and credit exceptions - Cutting list preparation and production gate compliance - Payment status monitoring (read-only for sales — posting is Cashier/Finance) - Refund request initiation - Staff purchase credit requests - Sales dashboard and KPI review

This SOP does **not** cover receipt confirmation (SOP-02), production execution (SOP-04/05), or refund approval/payment (SOP-02, Branch Manager procedures).

1.3 Roles and Responsibilities

Role	Responsibilities
Sales Officer (sales_staff)	Create customers, build quotations, attach cutting lists, monitor payment status, request refunds
Branch Manager (sales_manager)	All sales officer duties plus quotation clearance, order sign-off, production gate override, refund approval, credit exception approval (within limit)
MD (md)	Price exception approval, credit exceptions above BM limit, high-value refund approval

2. SYSTEM ACCESS AND NAVIGATION

2.1 Accessing the Sales Module

1. Sign in to Zarewa Online Office.
2. Confirm workspace branch in the top bar (e.g., Kaduna / BR-KD).
3. Click **Sales** in the sidebar — navigates to /sales.
4. Available tabs: **Quotations, Payments, Cutting list, Refunds, Customers.**

Permission required: sales.view minimum; quotations.manage for create/edit; customers.manage for customer CRUD.

Cashiers are excluded from the Sales module — they use Finance desk only.

2.2 Customer Dashboard

Navigate to /customers for the customer directory, or /customers/:customerId for individual customer dashboard showing: - Profile and contact details - All quotations with status and payment % - Ledger balance and transaction history - Advance deposits - Refund history - CRM interaction log

3. CUSTOMER MANAGEMENT

3.1 Creating a New Customer

When: Before first quotation for a new buyer.

Who: Sales Officer or Branch Manager.

Procedure:

1. Go to **Sales** → **Customers** tab (or /customers).
2. Click **New Customer**.
3. Complete required fields:
 - **Legal / trading name** — as it will appear on quotations and receipts
 - **Phone number** — primary deduplication key (normalised by system)
 - **Email** (optional) — secondary deduplication key
 - **Branch** — auto-set to current workspace branch
 - **Address** — delivery and correspondence
 - **Customer tier** — if pricing policy uses tiered pricing
 - **CRM tags** — optional categorisation
4. Click **Save**.
5. System checks for duplicate phone/email and warns if match found.
6. Customer receives internal ID; visible on dashboard.

Controls: - Phone normalisation via customerPhoneKey.js — enter in local format; system canonicalises - Duplicate customers cause quotation and ledger confusion — always search before creating - Customer data is branch-scoped; other branches cannot see unless org-wide read permission

3.2 Updating Customer Records

1. Open customer from list → Customer Dashboard.
2. Click **Edit** on profile section.
3. Update permitted fields.
4. **Sensitive edits** on customers with active quotations may require Edit Approval (6-digit code from BM/MD) — see SOP-10.

3.3 CRM Interactions

Log every significant customer touchpoint:

1. On Customer Dashboard → **Interactions** section.
2. Click **Log Interaction**.
3. Select kind: Call, Site visit, Follow-up, Complaint, Other.
4. Enter notes and date.
5. Optionally link to specific quotation reference.
6. Save — creates customer_crm_interactions record.

Best practice: Log interactions within 24 hours. Use for quotation follow-up alerts and MD customer intelligence in Command Centre.

3.4 Customer Search

- Use workspace search (/ → Search) for cross-entity lookup
 - Filter customer list by name, phone, tier
 - CEO role cannot access customer screens (executive summary only)
-

4. QUOTATION MANAGEMENT

4.1 Quotation Lifecycle Overview

draft → active (approved) → cutting_list_issued → in_production → completed
↓
cancelled (triggers refund if paid)

Status	Meaning	Who Triggers
Draft	Created, not yet confirmed	Sales staff
Active	Confirmed quotation — production eligible when paid	Sales / BM
Cutting list issued	CL generated and sent to factory	Sales / BM / Operations
In production	Factory currently producing	System on job start
Completed	All production and delivery confirmed	System on full delivery
Cancelled	Voided — refund workflow if payments received	BM / MD

4.2 Creating a Quotation

Procedure:

1. **Sales → Quotations → New Quotation.**
2. Select **customer** from dropdown (or create new customer first).
3. Set quotation header fields:
 - **Quotation date** — defaults to today
 - **Validity period** — system tracks expiry; follow-up alerts generated
 - **Payment terms** — Cash, Partial advance, Credit (requires credit exception)
 - **Delivery location** — if different from customer address
 - **Notes** — special instructions visible to operations
4. Add **line items** for each product:
 - **Category** — Long-span, Step-tile, Stone-coated, Accessory, Transport, Installation, Other
 - **Product / profile** — select from catalogue
 - **Gauge (mm)** — 0.18 to 0.55 standard range
 - **Colour** — from active colour register
 - **Quantity** — metres for sheets, units for accessories, sheets for stone
 - **Unit price (₹/m or ₹/unit)** — defaults from price list; editable
 - **Line total** — auto-calculated
5. Review **quotation total** at footer.
6. Click **Save** → system assigns ID e.g. QT-KD-26-0042.

Pricing validation: - System checks each line against **price list floor** (/price-list) and **material pricing workbook** - Price **at or above floor**: saves normally - Price **below floor**: save allowed with **warning banner**; cutting list and production **blocked** until MD approves price exception - Price snapshot captured at creation — subsequent price list changes do not affect saved quotation

4.3 Below-Floor Price Exception

When: Sales needs to offer price below approved floor (competitive deal, bulk discount, loyal customer).

Procedure:

1. Save quotation with below-floor lines (warning displayed).
2. Notify Branch Manager or MD with commercial justification.
3. MD opens quotation → **Price Exception** panel.
4. MD reviews lines, enters approval note, clicks **Approve Price Exception**.
5. API: PATCH /api/quotations/:id/md-price-exception-approve
6. Permission: md.price_exception.approve
7. Cutting list and production gates now pass for this quotation.

Audit: Approval recorded with MD user ID, timestamp, and note on quotation record.

4.4 Credit Exception

Default policy: All sales require advance or cash payment. Customer credit requires formal Credit Exception.

Procedure:

1. From quotation or customer dashboard → **Request Credit Exception.**
2. Enter:
 - Requested credit amount
 - Credit terms (days) — default 14, maximum 90
 - Commercial justification
3. Submit → status pending.
4. **Approval routing:**
 - Below BM limit (CREDIT_BRANCH_MANAGER_LIMIT_NGN): Branch Manager approves
 - Above BM limit: MD approval required
 - No policy configured: MD approval always required
5. On approval: status approved; delivery gate bypassed while exception active
6. Exception may be revoked or expired — system enforces expiry date

4.5 Quotation Follow-Up and Archiving

- System generates **follow-up alerts** for quotations approaching validity expiry
- **Archived** quotations are read-only; unarchive requires manager action
- Review stale quotations weekly on Sales dashboard

4.6 Quotation Clearance (Branch Manager)

Branch Manager reviews paid quotations for order sign-off:

1. Open **Manager Dashboard** (/manager) → **Order review** tab.
 2. Review quotations meeting payment threshold ($\geq 99.5\%$ treated as fully paid).
 3. Clear or flag quotations with notes.
 4. MD/admin may release payment holds and block refunds on specific quotations.
-

5. CUTTING LIST MANAGEMENT

5.1 Purpose

A Cutting List (CL) is the production instruction sent from Sales to the factory. It specifies exact metres per line, links to the quotation, and gates production start.

5.2 Prerequisites

Before creating a cutting list:

1. Quotation must be **active** (not draft or cancelled).
2. **Payment gate:** Customer must have paid $\geq$ cutting_list_min_paid_fraction of quotation value (default **70%**).
3. **Price gate:** No below-floor lines without MD price exception approval.
4. If payment below threshold: Branch Manager may grant **production gate override** with documented justification (≥ 10 char note).

5.3 Creating a Cutting List

Procedure:

1. Open quotation in **Sales – Quotations**.
2. Click **Create Cutting List** (or **Cutting list** tab).
3. System pre-fills lines from quotation.
4. Enter/adjust **metres per line**:
 - **Line type** — sheet, accessory, stone
 - **Length (m)** per piece
 - **Sheets** count where applicable
 - **Total metres** — auto-calculated
5. Verify **total metres** match commercial agreement.
6. Add accessory and stone lines if not already on quotation.
7. Click **Save** → assigns ID e.g. CL-KD-26-0015.
8. Status: cutting list issued; visible in Operations production queue.

Material readiness check: System may flag if coil stock insufficient for requested gauge/colour — coordinate with Operations before promising customer dates.

5.4 Cutting List Amendments

- Edits before production start: sales officer or BM
- Edits after production started: requires Edit Approval or manager unlock
- Metre changes after job completion: **production completion adjustments** (Operations) — see SOP-05

5.5 Production Gate Override (Branch Manager)

When customer has not met 70% payment but BM authorises production start:

1. BM opens **Manager Dashboard** → production gate queue.
2. Select quotation → **Approve Production** panel.
3. Enter justification note (minimum 10 characters).
4. Submit → records manager_production_approved_at_iso, approver, note, paid fraction at override.

5. Cutting list release now permitted.
 6. Override visible to MD in Executive Dashboard decision queue.
-

6. PAYMENT STATUS MONITORING

6.1 Sales Officer Role in Payments

Sales officers may post receipts if granted receipts.post permission. In many deployments, Cashier posts and confirms all receipts. Sales always has **read access** to payment status.

Payment status values (derived from receipts):

Status

Unpaid

Partial

Paid

6.2 Viewing Payment History

1. Open quotation → **Payments** section (blue history area).
2. Already-posted receipts are **read-only** — cannot edit in place.
3. Each receipt shows: amount, date, treasury account, clearance status (Pending/Cleared).
4. **Paid total** and **balance due** displayed prominently.

6.3 Posting a Receipt (if permitted)

See SOP-02 for full receipt posting procedure. Summary for sales:

1. Open quotation → **Record Payment**.
2. Enter **only new money** — do not re-enter historical receipts.
3. Amount, date, treasury account (cash/bank), reference text.
4. Amounts ≥ ₦100,000: type amount twice for confirmation.
5. Post → creates ledger entry and treasury movement (pending clearance).
6. Cashier must **confirm clearance** before receipt appears in financial reports.

Critical rule: Never mark a quotation as paid verbally or on paper without a system receipt. Production gates read system data only.

6.4 Advance Deposits

Customer pays before quotation exists:

1. Finance or Sales → **Advance deposit** against customer.
2. Creates ADVANCE_IN ledger entry.
3. When quotation created → **Link advance** to apply against quotation.

4. Creates ADVANCE_APPLIED entry; reduces balance due.
-

7. REFUND REQUESTS

7.1 When to Request a Refund

Category	Typical Cause
Order cancellation	Customer cancels before/during production
Unproduced meterage	Production not completed for ordered quantity
Overpayment	Customer paid more than invoice value
Transport issue	Delivery problem attributed to logistics
Installation issue	Defect discovered during/after installation
Accessory shortfall	Accessories not included with delivery
Stone flatsheet shortfall	Stone tiles short on delivery
Calculation error	Pricing or quantity mistake on quotation
Substitution difference	Gauge/profile substituted; price difference refunded
Customer commission	Sales agent commission credited to customer
Other	Catch-all; requires written justification

Minimum refundable balance: ₦1,000
(MIN_REFUND_QUOTATION_REMAINING_NGN)

7.2 Refund Request Procedure

Who: Sales Officer (refunds.request permission).

Procedure:

1. **Sales** → **Refunds** tab → **New Refund Request**.
2. Select **quotation** and **customer**.
3. Select **refund category** from list above.
4. Review **system-suggested lines** — starting points only; verify against evidence.
5. Adjust amounts per line with justification.
6. Enter **payee bank details** for transfer.
7. Review **production alignment warnings** (if any):
 - Order cancellation blocked if production completed (unless override note ≥10 chars)
 - Multi-category overlap requires acknowledgement
8. Click **Preview** → review calculated total.
9. Submit → status **Pending**.
10. Notify Branch Manager for approval.

Sales officer cannot: approve or pay refunds.

7.3 After Submission

- Track status on **Refunds** tab: Pending → Approved → Paid (or Rejected)
 - Respond promptly to approver queries
 - Gather evidence: photos, signed acknowledgements, site visit notes
 - Approved refunds appear in Cashier desk queue for payment — see SOP-02
-

8. STAFF PURCHASE CREDIT

8.1 Policy

Staff may purchase roofing materials on personal credit:

Rule

Minimum service for eligibility

Maximum outstanding obligation

Maximum per purchase

Maximum repayment period

Maximum concurrent active credits

Obligation account format

Repayment

8.2 Request Procedure

1. Staff member (or HR on behalf) → **Staff Purchase Credit Request**.
 2. Link to quotation for materials.
 3. BM endorses → creates obligation account.
 4. Repayment tracked via hr_payroll_line_loans and hr_staff_obligation_transactions.
-

9. PRICING ADMINISTRATION (READ ACCESS)

Sales staff have read access to current prices:

- **Price list** (/price-list) — floor prices per gauge/profile (MD manages)
- **Pricing policy** (/pricing-policy) — tier rules, customer price book
- Customer-facing price book: /api/pricing/customer-price-book.html

Sales officers must not alter price list — escalate to MD for changes.

10. SALES DASHBOARD AND KPIS

10.1 Sales Dashboard (/api/sales/dashboard/*)

KPI

Active quotations
Pipeline value
Collections (period)
Follow-up alerts
Payment gate queue

10.2 Branch Manager Pulse

On Manager Dashboard: - Produced sales progress vs org targets - Production metres progress - Open actions count

Targets configured: Settings → Governance → Org manager targets.

11. DAILY SALES OFFICE CHECKLIST

Sales Officer — Start of Day

- ☐ Confirm correct workspace branch
- ☐ Review quotation follow-up alerts
- ☐ Check payment status on quotations awaiting cutting list
- ☐ Clear CRM interaction backlog from previous day
- ☐ Review Manager dashboard for any returned/rejected items

Sales Officer — End of Day

- ☐ Log all customer interactions from today
- ☐ Update quotation notes for any verbal agreements
- ☐ Escalate blocked cutting lists (payment/price gate) to BM
- ☐ Confirm no draft quotations left incomplete with customers waiting

Branch Manager — Weekly

- ☐ Review order sign-off queue on Manager Dashboard
 - ☐ Clear stale quotations (archive or follow up)
 - ☐ Review credit exceptions expiring this week
 - ☐ Check refund pending queue — approve or reject within 48 hours
-

12. EXCEPTION HANDLING

Situation	Action
Customer phone duplicate warning	Search existing customer; merge if same entity; use different phone only if genuinely different person
Cutting list blocked — payment	Check receipt clearance status; request BM override if commercial

Cutting list blocked — price	decision made
Customer disputes quotation total	Escalate to MD for price exception
	Do not edit locked quotation; create credit note via refund or new quotation
System shows wrong paid amount	Ask Finance to run “Sync paid from receipts”; do not manually override
Refund rejected	Review approver notes; gather additional evidence; resubmit if appropriate
.	

13. TRAINING COMPETENCY CHECKLIST

New sales officer must demonstrate:

- ☐ Create customer without duplicate
 - ☐ Create quotation with 3+ line types
 - ☐ Identify below-floor warning and escalation path
 - ☐ Create cutting list on paid quotation
 - ☐ Read payment status and explain clearance workflow
 - ☐ Submit refund request with correct category
 - ☐ Log CRM interaction
 - ☐ Use Zare “Tour this page” on Sales module
-

End of SOP-01. Cross-references: SOP-02 (Cashier), SOP-04 (Operations), SOP-08 (Executive approvals).

SOP-02: CASHIER DESK

Zarewa Aluminium and Plastics Ltd — System of Operations v3.0

Department: Finance — Cashier Desk

System modules: /accounts (Desk, Receipts, Movements tabs)

Primary role: cashier

Supporting roles: finance_manager, md

1. PURPOSE AND SCOPE

1.1 Purpose

The Cashier Desk is the physical and system control point for all cash and bank money entering and leaving Zarewa at branch level. The cashier confirms that money recorded in the system matches money actually received or paid. This SOP ensures every receipt is cleared before entering financial reports, every payout is properly authorised, and daily cash balances reconcile to treasury accounts.

1.2 Scope

- Receipt confirmation and clearance
- Customer receipt posting (where cashier holds receipts.post)
- Advance deposit recording
- Bank deposit registration and allocation
- Refund payment execution (after approval)
- Payment request / expense payout execution
- Treasury transfers between accounts
- Staff obligation cash repayments
- End-of-day cash reconciliation

Out of scope: GL journals, bank reconciliation, period close (SOP-03 Accounting Desk).

1.3 Segregation of Duties

Cashier CAN

Post receipts

Confirm receipt clearance

Pay approved refunds

Execute treasury payouts

Request refunds

Process staff cash repayments

Critical: Refund requester ≠ approver ≠ payer (when ENFORCE_DUAL_CONTROL_PAYMENTS=1).

2. SYSTEM ACCESS

2.1 Navigation

1. Sign in → auto-routes to /accounts (Finance module).
2. Sidebar shows **Finance desk** (not “Sales” or “Accounting”).
3. Available tabs for cashier:
 - **Desk** — work queues (refunds awaiting payment, payouts, staff recovery)

- **Receipts** — confirm payment received
- **Movements** — fund transfers between treasury accounts

Accountant tabs (Payment register, Audit) are hidden or read-only for cashier role.

2.2 Permissions Required

Permission

cashier.desk.view
 cashier.receipts.confirm
 receipts.post
 finance.pay
 refunds.request
 treasury.manage
 ,

3. DAILY CASHIER ROUTINE

3.1 Opening Procedures (Start of Day)

1. Sign in; confirm workspace branch.
2. Open **Finance** → **Desk** tab.
3. Review overnight queue:
 - Refunds approved awaiting payment
 - Payment requests approved awaiting payout
 - Staff recovery items
4. Open **Finance** → **Receipts** tab.
5. Review **Pending clearance** receipts from previous day/sales postings.
6. Count physical cash in safe(s); note opening balance.
7. Compare to treasury cash account balance in system.
8. Report discrepancies to Branch Manager immediately — do not proceed with postings until resolved.

3.2 During the Day

- Process receipt confirmations as sales post new payments
- Execute approved payouts promptly (same day where possible)
- Register bank deposits as bank statement credits arrive
- Process treasury transfers as directed by Finance Manager

3.3 Closing Procedures (End of Day)

1. Confirm all physical receipts received today are posted and cleared in system.
2. Count closing cash balance in safe(s).
3. Compare to treasury cash account — must match.
4. Clear any remaining pending receipts or escalate with note.

5. Review Desk queue — no approved items should remain unpaid without documented reason.
 6. Report daily summary to Branch Manager if required by branch policy.
-

4. RECEIPT CONFIRMATION

4.1 Receipt Lifecycle

Sales/Cashier posts receipt → Pending clearance → Cashier confirms → Cleared → In reports/GL

Critical rule: Only **confirmed (Cleared)** receipts are recognised in ledger and financial reports. Unconfirmed receipts remain in cashier pending queue — invisible to management reports.

4.2 Posting a New Receipt

When: Customer payment received (cash, transfer, cheque, POS).

Procedure:

1. Obtain quotation reference or customer name from sales/customer.
2. Open quotation in **Finance → Receipts** (or Sales → Payments if cross-trained).
3. Review **blue history section** — already-posted receipts are read-only.
4. In editable section, enter **only today's new money**:
 - **Amount (₦)**
 - **Payment date**
 - **Treasury account** — select correct cash safe or bank account
 - **Reference text** — transfer ref, POS slip number, cheque number, depositor name
5. **High-value control:** Amounts ≥ ₦100,000 require typing the amount twice.
6. Click **Post Receipt**.
7. System creates:
 - Ledger entry (type RECEIPT)
 - Sales receipt record (status: Pending clearance)
 - Treasury movement (pending)
8. Note receipt ID (e.g., RCP-KD-26-0088) for customer acknowledgement.

Duplicate detection: System blocks same quotation/amount/date combination. Override only with valid reason — audit captures override.

4.3 Confirming Receipt Clearance

When: Bank/cash evidence verified — money is physically in account.

Procedure:

1. **Finance** → **Receipts** tab.
2. Filter or locate **Pending clearance** items.
3. For each receipt:
 - Verify amount matches bank alert, deposit slip, or cash count
 - Verify customer and quotation reference
 - Verify treasury account selection was correct
4. Click **Confirm Payment Received**.
5. API: PATCH /api/sales-receipts/:id/bank-confirmation with { confirmed: true }
6. Status changes to **Cleared**.
7. Treasury movement finalised; GL auto-posted (AP1c):
 - Dr Cash/Bank (1000)
 - Cr Accounts Receivable

If evidence does not match: - Do not confirm - Contact sales officer or customer for clarification - If wrong posting: escalate to Finance Manager for reversal (SOP-03) — never edit cleared receipt

4.4 Receipt Clearance and Refunds

Refunds on a quotation are blocked until all receipts on that quote are Cleared.

Before processing refund payment, verify all quotation receipts show Cleared status.

5. ADVANCE DEPOSITS

5.1 Customer Advance (Payment Before Quotation)

Procedure:

1. Customer pays without active quotation.
2. **Finance** → select customer → **Record Advance**.
3. Enter amount, date, treasury account, reference.
4. Post → creates ADVANCE_IN ledger entry.
5. Customer advance balance visible on customer dashboard.

5.2 Applying Advance to Quotation

1. When quotation created, open **Link Advance** on quotation or customer dashboard.
2. Select advance deposit(s) to apply.
3. System creates ADVANCE_APPLIED entries; reduces quotation balance due.

5.3 Refunding Advance

If customer requests return of unapplied advance:

1. Finance Manager approves advance refund.

2. Cashier executes via treasury payout.
 3. Creates ADVANCE_REFUND or equivalent ledger entry.
-

6. BANK DEPOSIT WORKFLOW

6.1 Purpose

Unlinked bank credits on the bank statement (not yet tied to a customer receipt) are registered in the Bank Deposit pool, then allocated to their source.

6.2 Registering a Bank Deposit

1. Bank statement shows credit not matching any receipt.
2. **Finance** → Bank deposit register (or Accounting desk for complex cases).
3. Create new deposit:
 - Amount, date, bank account
 - Description from bank statement
4. Assigns ID: BD-KD-26-0003
5. Status: **OPEN**

6.3 Allocating Deposits

Status	Meaning
OPEN	Registered, not linked
RESERVED	Temporarily reserved (30-min window)
PARTIAL	Partially linked
ALLOCATED	Fully linked to receipts/advances
RECLASSIFIED	Non-sales: expense_offset, inter_branch, refund_out, other_income
REVERSED	Bank error or returned transfer

Procedure to allocate:

1. Open deposit → **Allocate**.
 2. Link to customer receipt(s) or advance deposit(s).
 3. Confirm allocation amounts sum to deposit total.
 4. Status → **ALLOCATED**.
-

7. REFUND PAYMENT

7.1 Prerequisites

Before paying any refund:

- ☐ Refund status = **Approved**

- ☐ Approver ≠ current cashier (dual control)
- ☐ All quotation receipts **Cleared**
- ☐ Payee bank details verified
- ☐ Sufficient balance in paying treasury account

7.2 Payment Procedure

1. **Finance** → **Desk** tab → **Refunds awaiting payment** queue.
2. Select approved refund.
3. Review: customer, amount, payee bank, approver name.
4. Verify bank balance sufficient.
5. Click **Pay Refund**.
6. API: POST /api/refunds/:id/pay
7. System creates treasury movement (source: REFUND_PAYOUT).
8. Refund status → **Paid**.
9. Execute actual bank transfer using bank portal.
10. File transfer confirmation with refund record.

High-value refunds (>₦1,000,000): Verify MD approval recorded before payment.

8. PAYMENT REQUEST / EXPENSE PAYOUT

8.1 Flow

Staff creates expense request → BM/MD approves → Cashier pays from Desk

8.2 Payment Procedure

1. **Finance** → **Desk** tab → **Approved payment requests**.
 2. Review: payee, amount, expense category, approver.
 3. Verify approval within authority (BM ≤ ₦200k; MD above).
 4. Click **Pay** → treasury movement created.
 5. Execute bank transfer or cash disbursement.
 6. Obtain payee signature on payment voucher where required.
-

9. TREASURY TRANSFERS

9.1 Inter-Account Transfers

Move funds between treasury accounts (e.g., cash safe → bank deposit):

1. **Finance** → **Movements** tab.
2. **New Transfer:**
 - From account
 - To account

- Amount, date, reference
- 3. Post → creates paired treasury movements.
- 4. Both accounts update immediately.

9.2 Staff Obligation Cash Repayment

Staff repays loan/purchase credit in cash:

1. Look up staff obligation account (OBL-XX-YY-NNNN).
 2. Record cash repayment at cashier.
 3. Dr Cash/Bank, Cr Staff Obligation Ledger.
 4. Reduces outstanding obligation balance.
-

10. CONTROLS AND AUDIT

10.1 Key Controls

Control

Receipt clearance mandatory

High-value double entry

Duplicate receipt block

Period lock

Refund payer ≠ approver

Zero amount block

CSRF on all writes

10.2 Audit Trail

All cashier actions logged: - receipt.post — new receipt - receipt.bank_confirmation — clearance - refund.pay — refund payout - treasury.transfer — inter-account move - payment_request.pay — expense payout

Review: Settings → Audit log (admin) or Finance → Audit tab (accountant).

11. ERROR CORRECTION

Cashier does not reverse receipts. Escalate to Finance Manager:

1. Identify error (wrong amount, wrong customer, duplicate).
2. Document evidence (bank statement, customer confirmation).
3. Finance Manager executes **reversal** → POST /api/ledger/reverse-receipt.
4. Verify treasury net corrected.
5. Re-post correct receipt if needed.
6. Never delete or overwrite cleared receipts.

12. DAILY CHECKLIST

Opening

- ☐ Branch confirmed
- ☐ Desk queue reviewed
- ☐ Pending receipts from yesterday identified
- ☐ Physical cash count matches treasury opening balance

During Day

- ☐ Each customer payment confirmed same day where possible
- ☐ Approved refunds paid within 24 hours
- ☐ Bank deposits registered as credits appear

Closing

- ☐ All today's receipts cleared or escalated
 - ☐ Physical cash matches treasury closing balance
 - ☐ No approved payouts left in queue without reason
 - ☐ Discrepancies reported to BM
-

13. TRAINING COMPETENCY

- ☐ Post receipt with double-entry for ₦100k+
 - ☐ Confirm clearance on pending receipt
 - ☐ Pay approved refund from Desk queue
 - ☐ Register and allocate bank deposit
 - ☐ Execute treasury transfer
 - ☐ Explain why unconfirmed receipts don't appear in reports
 - ☐ Demonstrate escalation path for receipt error
-

End of SOP-02. Cross-references: SOP-01 (Sales), SOP-03 (Accounting), SOP-08 (MD approvals).

SOP-03: ACCOUNTING DESK

Zarewa Aluminium and Plastics Ltd — System of Operations v3.0

Department: Finance — Accounting Desk

System modules: /accounting, /accounts (accountant tabs), /reports

Primary role: finance_manager (Accountant / Head of Accounts)

Supporting roles: md, admin

1. PURPOSE AND SCOPE

The Accounting Desk is Zarewa's financial control centre. It manages the General Ledger, bank reconciliation, accounts payable and receivable sub-ledgers, fixed assets, accounting period locks, and the three-phase accounting policy engine (AP1c, AP2, AP3). This SOP covers month-end close, GL operations, and management reporting.

Segregation: Accountants reconcile and journal; Cashiers receive and pay. Accountants hold finance.cross_branch_post for controlled cross-branch ledger entries.

2. ACCOUNTING DESK NAVIGATION

Route: /accounting

Zones: Home · Close · Reports · Registers · Policy · Operations

Tab

Overview
Opening Pack
Month-end close
Reconciliation
Statements / GL
Creditors / Debtors / Assets
Inter-branch
Deposit policy (AP1c)
Supplier AP (AP2)
Production costing (AP3)
Branch P&L
Credit approval
Payroll
,

3. GENERAL LEDGER STRUCTURE

3.1 Chart of Accounts (Key Codes)

Code	Account	Type
1000	Cash / Bank	Asset
1200	Staff Loans Receivable	Asset
1300	Inventory / Closing Stock	Asset

1500–1504	Fixed Assets (Plant, Land, Furniture, Generator)	Fixed Asset
2000	Accounts Payable — Suppliers	Liability
2400	Pension Payable	Liability
2500	Customer Advances / Refunds Payable	Liability
3200	Chairman Withdrawal (Equity Draw)	Equity
4000	Sales Revenue	Income
5010–5050	COGS (Fuel, Maintenance, Corrugation, Production, Purchases)	COGS
6000–6400	Operating Expenses	OpEx
6100	Depreciation	OpEx

3.2 Expense Category → GL Mapping

Category	GL	Notes
Wages	6000	Direct production wages
Fuel & lubricant	5010	
Maintenance	5020	
Outside corrugation	5030	
Production cost / Accessories	5040	
Purchases / Carriage inward	5050	
Admin / Office / HQ / Welfare / Security / Others	6120	Others requires 40-char justification
Admin salary	6110	
Pension	2400	Liability — not expense
Staff loan	1200	Asset — not expense
Chairman withdrawal	3200	Equity draw — not expense
Capex categories	1500–1504	Auto-registers fixed asset on full payment

3.3 “Others” Expense Controls

Control

Minimum justification
Finance review flag
AP3 unclassified alert

Branch coaching alert

4. ACCOUNTING PHASES (AP1c → AP2 → AP3)

4.1 AP1c — Production Recognition

Policy V1: Revenue recognised at **production completion**, not delivery or invoice date.

On production job completion, system auto-posts: - Dr Accounts Receivable (customer ledger) - Cr Sales Revenue (4000)

On receipt clearance: - Dr Cash/Bank (1000) - Cr Accounts Receivable

Dry-run: Accounting desk → AP1c → run dry-run before go-live.

4.2 AP2 — Received-Basis AP + Inventory

- Payables recognised at **GRN** (not payment date)
- Inventory valued at **landed cost** (including carriage inward)
- Supplier advances tracked; applied on GRN
- Diagnostics: /accounting → AP2 → supplier advance report

4.3 AP3 — Branch P&L + Material Costing

- Standard vs actual material cost per production job
- Payroll labour allocated per job
- Branch P&L draft report
- Costing readiness check before relying on branch margin reports

Migration rule: AP3 requires AP2 requires AP1c. Coordinate with MD before phase activation.

5. BANK RECONCILIATION

5.1 Procedure

1. **Accounting** → Reconciliation tab (or /accounts/bank-reconciliation redirect).
2. Import bank statement:
 - **CSV:** bankDateISO,description,amountNgn with optional header
 - **JSON:** up to 500 lines per batch via POST /api/bank-reconciliation/import
3. System fingerprints lines (bankReconFingerprint.js) to detect duplicates.
4. Match each line to:
 - Customer receipts (cleared)
 - Expense payments
 - Supplier payments
 - Inter-branch transfers

5. Mark unmatched lines as **Review** for investigation.
6. Weekly: Finance lead reviews all Review lines.
7. Month-end: zero unmatched lines or documented carry-forward.

5.2 Control Tie-Out

Run control tie-out report: GL cash balance must equal reconciled bank balance.
Discrepancies block period lock.

6. PAYABLES RECONCILIATION

Three-way match (month-end):

Leg

Ordered

Received

Paid

1. Run purchase register report (by received date).
 2. Compare AP2 supplier diagnostics for unmatched advances.
 3. Resolve short receipts with Procurement (MD notified on coil shorts).
-

7. RECEIVABLES AND WRITE-OFFS

7.1 Debtors Register

Accounting desk → Creditors/Debtors → Debtors (AR) sub-ledger.

Review aging: outstanding balances, days overdue.

7.2 Write-Off Policy

Receivable write-offs above configured threshold require MD approval (receivableWriteOffPolicy.js).

Procedure: 1. Document customer insolvency, legal pursuit exhaustion, or board decision. 2. MD approves write-off on quotation/customer record. 3. Post adjusting journal or use register settlement workflow. 4. Audit trail mandatory.

8. FIXED ASSETS AND DEPRECIATION

8.1 Asset Creation

- **Manual:** Finance creates in Accounting → Assets
- **Automatic:** Capex expense fully paid → fixedAssetAutomationOps.js registers asset

Category

Plant & machinery

Land & buildings

Furniture & fittings

Generator

8.2 Depreciation Run

Monthly (month-end): 1. Run depreciation via depreciationRunOps.js 2. Posts: Dr Depreciation (6100), Cr Accumulated Depreciation 3. Verify in trial balance before period lock

8.3 Disposal

1. Finance initiates disposal with sale/scrap value
 2. GL: Dr Accumulated Depreciation + Dr Cash, Cr Asset Cost, Cr/Dr Gain/Loss
 3. Asset removed from active register
-

9. INTER-BRANCH LOANS

Cross-branch cash funding recorded as inter-branch loans (not expense):

1. Finance proposes loan: lending branch → borrowing branch
 2. MD approves (inter_branch_loan.md_approve)
 3. Status: proposed → md_approved → executed
 4. Repayments tracked until settlement
 5. Both branches see loan on treasury reports
-

10. REGISTER SETTLEMENTS

Creditor/debtor register lines follow: propose → review → decision → pay

1. Open register line in Accounting desk
 2. Propose settlement amount and method
 3. Reviewer confirms
 4. Finance executes payment
 5. Each step tracked in accounting_register_settlements
-

11. PAYROLL FINANCE HANDOFF

After HR payroll chain complete:

1. Verify payroll run **locked** with MD approval (md_approved_at_iso recorded)
2. **Accounting** → Payroll tab

3. Export bulk bank file (hr.payroll.export)
4. Execute bank transfers
5. Post treasury movements
6. GL journal CSV: salary expense, PAYE, pension, net pay

Gate: Payroll cannot lock without GM HR **or** MD approval; cannot pay without lock.

12. MONTH-END CLOSE CHECKLIST

Step

- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9

Period lock: assertPeriodOpen() blocks all back-dated posts to locked YYYY-MM.

13. MANAGEMENT REPORTS

Route: /reports (requires reports.view)

Report

Receipts register

AR as-at

Sales bridge

Expenses pack

Refunds pack

Purchase register

Stock coil as-at

Governance pack

MD operations pack

Daily / weekly pack

14. RECEIPT CORRECTIONS (FINANCE AUTHORITY)

1. Identify error from exception queue
2. Gather bank/cash evidence
3. **Reverse** mistaken posting — POST /api/ledger/reverse-receipt
4. Verify treasury net corrected
5. **Re-post** true amount only
6. Confirm quotation paidNgn syncs via syncQuotationPaidFromReceipts
7. Close queue item with audit note

Split correction: POST /api/ledger/correct-receipt-split for allocation errors.

15. DAILY / WEEKLY CADENCE

Daily: Clear payment exception queue; review new uncleared receipts aging >24h.

Weekly: Bank reconciliation Review lines; aged exception report (>7 days) to Finance lead.

Monthly: Full close checklist \$12.

End of SOP-03. Cross-references: SOP-02 (Cashier), SOP-06 (Procurement AP), SOP-07 (Payroll).

SOP-04: OPERATIONS & STORE

Zarewa Aluminium and Plastics Ltd — System of Operations v3.0

Department: Operations / Store

System modules: /operations, /operations/coils/:coilNo, /operations/material-exceptions

Primary role: operations_officer (Storekeeper)

Supporting roles: sales_manager, md

1. PURPOSE AND SCOPE

Operations and Store is the authoritative source for physical inventory truth: coil register, accessory stock, stone inventory, goods receipt (GRN), customer deliveries, and material exceptions. Sales sees status read-only; Operations confirms what was produced and delivered.

2. OPERATIONS MODULE NAVIGATION

Route: /operations

Tab

Overview

Stock management

Material exceptions

Production line

3. GOODS RECEIPT (GRN)

3.1 Prerequisites

- Approved PO in status **On loading**, **In Transit**, or **Approved** (per local policy)
- Physical goods arrived at branch
- Weighbridge ticket (coil) or packing list (accessories/stone)

3.2 Coil GRN Procedure

1. **Operations – Stock management – Post GRN.**
2. Select PO and PO line.
3. Enter received data:
 - **Coil number** (manufacturer serial — unique)
 - **Gross weight (kg)**
 - **Gauge, colour, profile, material type** (aluzinc/aluminium)
 - **Received date**
4. System creates coil_lots record; updates qty_remaining.
5. GL (AP2): Dr Inventory (1300), Cr AP (2000).
6. If short-receipt vs PO qty: system notifies MD (notifyMdCoilShortReceipt).

3.3 Accessory / Stone GRN

1. Select PO line (accessory or stone type).
2. Enter quantity received vs PO quantity.
3. Updates products branch stock (stock_level).
4. Stone: track sheets; system converts sheets to m² for production.

3.4 GRN Controls

- Cannot GRN against cancelled PO
 - Short receipt tolerance configurable; above tolerance – MD work item
 - Landed cost includes carriage inward from PO transport fields
-

4. COIL REGISTER MANAGEMENT

4.1 Coil Lot Attributes

Field

Coil number
qty_remaining
qty_reserved
Gauge, colour, profile
current_status
production_blocked

4.2 Coil Profile

Navigate: /operations/coils/:coilNo

Shows: full traceability, control events (CREV), production allocations, reservation history.

4.3 Coil Control Events (CREV)

Every coil movement logged immutably:

Event Type

Return inward
Return outward
Head trim
Supplier defect
Split
Scrap
Finish roll

Tail finish: When remaining ≤ 85 kg, system flags “Roll Finished” and clears residual.

4.4 Coil Requests

Store requests coil from yard/register: 1. Create **Coil Request** (CREQ-XX-YY-NNNN) 2. Approve and fulfil 3. Stock movement recorded

5. STOCK MANAGEMENT

5.1 Finished Goods (FG)

Post-production stock tracked in products by gauge, colour, type per branch.

5.2 Low Stock Alerts

Dashboard KPI: SKUs below threshold. Review daily on Overview tab.

5.3 Stock Adjustments

Requires inventory.adjust permission. Document reason; manager approval for material adjustments.

5.4 In-Transit Loads

Factory-to-branch goods in transit: IT-KD-26-NNNN. Reconciled at month-end stock register.

6. DELIVERIES

6.1 Authority

Operations confirms delivery — not Sales. Sales sees delivery status read-only.

6.2 Delivery Procedure

1. FG ready from completed production job.
2. **Operations** → create **Delivery** (DLV-XX-YY-NNNN).
3. Link to quotation and delivery lines (metres per product).
4. **Delivery gate check:**
 - DELIVERY_PAYMENT_GATE=enforce: blocked if balance due and no credit exception
 - warn: warning shown but proceeds
 - off: no check
5. Dispatch goods; customer signs delivery note (POD).
6. **Confirm delivery** in system with POD reference.
7. Stock deducted; quotation delivery status updated.

6.3 Delivery Credit

If dispatch before full payment required: - Active **credit exception** on customer, OR - Branch Manager **delivery credit** approval on Manager dashboard

7. MATERIAL EXCEPTIONS

See dedicated workflow — summary:

1. **New incident** → type, coil/job links, kg/metres, storekeeper + operator names
2. **Save draft** → print with draft watermark if needed
3. **Submit** → Branch Manager queue

4. **Approve & post** → stock adjusted; metres to offcut pool
5. **Void** (not delete) with manager reason if error

Types: coil_damage, production_offcut, missing_asset, asset_loss

Full detail: backend docs/MATERIAL_EXCEPTIONS_SOP.md and SOP-05 §6.

8. MONTHLY STOCK REGISTER — 4-STAGE SIGN-OFF

Stage

- 1
- 2
- 3
- 4

Locked register: No further edits; snapshot in work_item_print_snapshots.

Capture via **Reports** → Stock register month-end.

9. MATERIAL REQUESTS (INTER-BRANCH)

Branch requests stock from another branch:

1. Create **Material Request** (MREQ-XX-YY-NNNN)
 2. Fulfilling branch approves and dispatches
 3. Stock movements on both branches
 4. May create inter-branch treasury loan if applicable
-

10. DAILY OPERATIONS CHECKLIST

Opening: - ☐ Branch confirmed - ☐ Review production queue - ☐ Check coil availability for today's cutting lists - ☐ Review in-transit loads expected today

During day: - ☐ GRN all arrivals same day - ☐ Register new coil lots immediately after weighbridge - ☐ Log material incidents before end of shift

Closing: - ☐ Complete open GRNs - ☐ Confirm deliveries dispatched are confirmed in system - ☐ Report discrepancies to BM

11. CONTROLS

- Coil movements audited via coil_control_events
- No delete on material incidents — void only

- GRN creates AP liability under AP2 — coordinate with Finance
 - Operations is authoritative for produced metres and delivery truth
-

End of SOP-04. Cross-references: SOP-05 (Production), SOP-06 (Procurement GRN link), SOP-01 (Cutting lists).

SOP-05: PRODUCTION

Zarewa Aluminium and Plastics Ltd — System of Operations v3.0

Department: Production

System modules: /operations — Production line, Live Production Monitor

Primary roles: operations_officer, Production Manager (desig_pm), QCO (desig_qco)

Supporting roles: sales_manager (QC sign-off)

1. PURPOSE AND SCOPE

Production transforms raw coil into finished roofing metres per cutting list instructions. This SOP covers production job lifecycle, coil allocation, conversion quality control, WIP accounting, revenue recognition trigger (AP1c), and post-completion adjustments.

2. PRODUCTION WORKFLOW OVERVIEW

Cutting List issued → Production Job created (Planned)
→ Coil allocated → Job Started (Running)
→ Metres recorded → Conversion check
→ Job Completed → Revenue recognised (AP1c)
→ Delivery created → Customer POD

Job Status

Planned

Running

Completed

Cancelled

3. RELEASING TO PRODUCTION QUEUE

3.1 Prerequisites

- ☐ Cutting list exists and status = issued
- ☐ Payment gate passed OR BM production override recorded

- ☐ Price exception approved (if below-floor quotation)
- ☐ Coil stock available for gauge/colour

3.2 Procedure

1. **Operations** → **Production line** → **Production queue**.
 2. Select cutting list / quotation.
 3. Click **Release to production**.
 4. System creates Production Job PJ-KD-26-NNNN (status: Planned).
 5. Job appears in Live Production Monitor.
-

4. COIL ALLOCATION

4.1 Procedure

1. Open production job.
2. **Allocate coil(s)** from available coil lots:
 - Match gauge, colour, profile to cutting list requirements
 - Enter kg to allocate per coil
3. System reserves kg on coil_lots.qty_reserved.
4. Logs coil control event (CREV) — return outward to floor.
5. Links recorded in production_job_coils.

4.2 Blocked Coils

If production_blocked=true on coil lot: - Cannot allocate until block reason resolved - MD auto-notified on coil short-receipt blocks

4.3 Coil Consumption

Consumption follows actual coil tag weights through production. On completion, remaining kg returned via control event or finish-roll if ≤85 kg.

5. STARTING AND RUNNING PRODUCTION

5.1 Start Job

1. Open job in Live Production Monitor.
2. Click **Start production**.
3. Status → **Running**.
4. WIP balance opened (AP3): coil cost debited to WIP.

5.2 During Production

- Record progress metres per line
- Log material issues via **Report material issue** (quick-create material incident)
- Monitor coil remaining weight

6. COMPLETING PRODUCTION

6.1 Completion Procedure

1. Measure actual output metres per line.
2. Enter **produced metres** on job completion form.
3. System computes **conversion variance** vs theoretical (standard kg/m per gauge).
4. If variance **>5%**: flagged for manager review.
5. Select **conversion reason** if band = High or Low:
 - Material defect, coil splicing, cutting error, specification change, calibration difference
6. Click **Complete job**.
7. Status – **Completed**.

6.2 Conversion QC Bands

Band

OK

High

Low

Pending

BM QC sign-off: Manager Dashboard → Production QC tab.

6.3 Offcut Usage

If production uses material from incident offcut pool: - Select incident(s) on completion form - Completion shows “supplied from offcut” - Pool metres decremented

6.4 Accessory and Stone Usage

- production_job_accessory_usage — accessory fulfillment per job
 - production_job_stone_flatsheet_usage — stone sheets consumed
 - Tracked separately from sheet metres
-

7. REVENUE RECOGNITION (AP1c)

Policy V1 — critical accounting rule:

Revenue is recognised at **production completion**, not delivery or invoice date.

On job completion, system auto-posts GL: - **Dr** Accounts Receivable (customer ledger) - **Cr** Sales Revenue (GL 4000)

Implication: Customer balance shows revenue on books before physical delivery. Finance and Sales must understand this for customer statements and refund calculations.

8. WIP AND COSTING (AP3)

Event

Job start

Job complete

Accessory usage

AP3 reports: standard vs actual cost per metre; costing readiness check on Accounting desk.

9. PRODUCTION COMPLETION ADJUSTMENTS

Post-completion metre corrections (customer dispute, measurement error):

1. Operations or BM initiates adjustment
 2. Documents reason (≥ 10 chars)
 3. Adjusts produced metres on job
 4. May trigger revenue/AR adjustment via AP1c rules
 5. Audited in `production_completion_adjustments`
-

10. JOB INTELLIGENCE PANEL

API: GET `/api/production-jobs/:id/intel`

Displays: - Conversion alert band - Planned vs actual metre variance % - Stone/accessory rollup - Quote paid % and BM production-gate override status - Refund risk flags

Available in Live Production Monitor — collapsible panel.

11. PRODUCTION PAYMENT GATE OVERRIDE

When quotation below 70% paid but BM authorises production:

1. BM → Manager Dashboard → Production gate
 2. **Approve production** with note (≥ 10 chars)
 3. Records: `manager_production_approved_by_*`, note, paid fraction at override
 4. Also available: **ProductionPaymentGateOverridePanel** in Live Production Monitor
-

12. QUALITY CONTROL (QCO)

Quality Control Officer (`desig_qco`) responsibilities:

- Sign off on High/Low conversion bands
 - Verify gauge and colour match cutting list
 - Reject completion if specification deviation — return to Running
 - Link QC flag to production job record
-

13. PRODUCTION DASHBOARD KPIS

KPI

Metres produced (MTD)

Jobs in queue

Conversion QC gaps

Payment gate breaches

14. DAILY PRODUCTION CHECKLIST

- ☐ Release new cutting lists to queue (morning)
 - ☐ Allocate coils before start
 - ☐ Complete jobs same day where possible
 - ☐ QC sign-off on all High/Low completions before close
 - ☐ Report material incidents for scrap/damage
 - ☐ Coordinate deliveries with Operations store
-

End of SOP-05. Cross-references: SOP-04 (Store/GRN), SOP-03 (AP1c revenue), SOP-01 (Cutting lists).

SOP-06: PROCUREMENT

Zarewa Aluminium and Plastics Ltd — System of Operations v3.0

Department: Procurement

System modules: /procurement, /procurement/suppliers/:id, /procurement/transport-agents/:id

Primary roles: md, Procurement Officer (desig_po, desig_hop)

Supporting roles: operations_officer (GRN), finance_manager (payment)

1. PURPOSE AND SCOPE

Procurement manages the acquisition of raw materials (coil, stone, accessories), transport haulage, and supplier relationships. At Zarewa, procurement is centralised at HQ under MD authority. This SOP covers the complete PO lifecycle from requisition through payment.

2. PROCUREMENT MODULE

Route: /procurement

Tab

Purchases

Payments

Transport catch-up

Suppliers

Conversion

3. SUPPLIER MANAGEMENT

3.1 Scope

Suppliers are **company-wide** master data — not branch-scoped. Shared across Kaduna, Yola, Maiduguri.

3.2 Registering a Supplier

1. **Procurement → Suppliers → New Supplier.**
2. Complete:
 - Legal name, trading name
 - Tax ID (deduplication key)
 - Phone (normalised)
 - Contact persons
 - Bank account details (multiple accounts supported)
 - Attachments (contracts, certificates)
3. Save → supplier profile at /procurement/suppliers/:supplierId.

3.3 Transport Agents

Haulage partners managed separately at /procurement/transport-agents/:agentId.
Linked to PO transport and in-transit loads.

4. PURCHASE ORDER TYPES

Derived from line items (poLineTypes.js):

PO Kind

Coil

Accessories

Stone/flatsheet
Supplier advance
Mixed
,

5. PO LIFECYCLE

5.1 Status Flow

Draft → Pending → Approved → On loading → In Transit → Received
 ↘ Rejected

Status

Pending
Approved
On loading
In Transit
Received

5.2 Creating a Purchase Order

1. **Procurement** → **Purchases** → **New PO**.
2. Select **supplier**.
3. Select PO type modal: **Coil** / **Stone** / **Accessory**.
4. Add lines:
 - **Coil**: gauge, colour, kg, price per kg, expected coil specs
 - **Stone**: sheets, m², product type
 - **Accessory**: SKU, quantity, unit price
5. Enter transport fields: agent, haulage cost, expected delivery date.
6. Save → PO-KD-26-NNNN (branch code of creating workspace).
7. Status: Pending (or Approved if creator has authority).

5.3 PO Approval

- MD approves major coil POs (default policy)
- BM may approve within local limits if configured
- Rejected POs cannot proceed to loading

5.4 Transport and In-Transit

1. Assign **transport agent** on approved PO.
2. Update status → **On loading** when loading commences.
3. Post **haulage payment** (optional) → may advance to **In Transit**.
4. syncPurchaseOrderTransportPaymentState links treasury payment to status.
5. Create **In-Transit Load** (IT-KD-26-NNNN) for tracking.

5.5 Goods Receipt (Handoff to Operations)

When goods arrive at branch — **Operations posts GRN** (SOP-04 §3): - PO status → **Received** when all lines fully received - Short receipt: tolerance check; above tolerance → MD work item (procurementWorkItems.js)

6. SUPPLIER PAYMENT

6.1 Procedure

1. **Procurement** → **Payments** tab (or Finance desk).
2. Select PO / payable line.
3. Verify three-way match readiness: ordered, received, due.
4. Finance posts **supplier payment**:
 - Links to PO and treasury account
 - Updates accounts_payable and supplier_paid_ngn
5. AP2: payment clears AP liability created at GRN.

6.2 Supplier Advances (AP2)

Pre-payment before goods receipt: 1. Record advance on PO 2. Tracked as supplier advance under AP2 3. On GRN: advance applied against payable 4. Diagnostics: Accounting desk → AP2 → unmatched advances

7. MATERIAL REQUESTS

Internal branch-to-branch or store requests:

1. Create **Material Request** (MREQ-XX-YY-NNNN) from Office or Operations
 2. Approve and fulfil
 3. Stock movements; optional inter-branch treasury entries
-

8. CONVERSION STANDARDS

Procurement → **Conversion** tab:

- Coil density (kg/m) per gauge — used for conversion variance calculation
- Price list administration (also /price-list)
- Material pricing workbook floors

Changes affect production conversion checks — coordinate with Production Manager before updating.

9. PROCUREMENT DASHBOARD

/api/procurement/dashboard/*: - Open POs by status - In-transit loads - Supplier payable aging - Short receipt alerts

10. CONTROLS

Control	Detail
MD notification	Coil short receipt above tolerance
Company-wide suppliers	Single supplier record prevents duplicate payments
PO edit after Received	Requires Edit Approval
Transport payment sync	Status auto-updates on haulage treasury post
AP2 liability	GRN creates AP; payment clears — not payment-created liability

11. MONTH-END PROCUREMENT CHECKLIST

- ☐ All received goods GRN'd
 - ☐ In-transit loads reconciled
 - ☐ Supplier payables match GRN register
 - ☐ Unmatched advances investigated (AP2 diagnostics)
 - ☐ Open POs documented (on order vs in transit)
-

End of SOP-06. Cross-references: SOP-04 (GRN), SOP-03 (AP2 payables), SOP-08 (MD approvals).

SOP-07: HUMAN RESOURCES

Zarewa Aluminium and Plastics Ltd — System of Operations v3.0

Department: Human Resources

System modules: /hr/*, /team-hr/*, /executive-hr/*, /my-profile/*

Primary roles: hr_admin, gmhr

All staff: self-service via /my-profile

1. PURPOSE AND SCOPE

HR manages the complete employee lifecycle: onboarding, time & absence, payroll, loans, discipline, transfers, separations, and executive benefits. The ERP enforces multi-stage approval chains and blocks account deactivation until exit clearance completes.

2. HR MODULE STRUCTURE

Hub

HR Dashboard

Employees

Time & Absence

Payroll & credit

Talent

Cases & exit

Documents

Analytics

Administration

Team HR

Executive HR

Self-Service

3. EMPLOYEE ONBOARDING

3.1 Staff Master File — Required Fields

Assessed by `assessStaffFileCompleteness()`:

- Full name, date of birth, phone
- Branch / site, job title / designation
- Date joined, probation end date (permanent staff)
- Bank account details
- Next of kin (name + phone)
- Highest academic qualification

3.2 Registration Procedure

1. **HR → Employees → Register Employee.**
2. Complete all required fields.
3. Assign designation, salary level (L1–L7), branch.
4. Create system login (Settings → Team & access) with appropriate role.
5. Set `selfServiceEligible=true` for My HR access.

6. Probation tracked — alert when approaching end (default 6 months).

3.3 Staff Numbering

Human-readable staff numbers assigned via hrStaffNumbering.js per branch policy.

4. LEAVE MANAGEMENT

4.1 Leave Types

Type

Annual

Sick

Maternity

Compassionate

Leave without pay

Other

4.2 Leave Request Workflow

Employee draft → submit → hr_review → branch_manager_review → gm_hr_review
→ approved

↘ rejected (at any stage)

Step

Submit

HR review

Branch endorse

GM HR final

4.3 Employee Procedure

1. **My HR → Time off → Request leave.**
2. Select type, dates, reason.
3. Submit → track status on **My requests**.

4.4 Team HR Endorsement (Branch Manager)

1. **Team HR → Time & absence.**
 2. Review team member requests.
 3. Endorse or return with comments.
-

5. LOAN MANAGEMENT

5.1 Staff Loan Workflow

Same chain as leave: submit → HR review → branch endorse → GM HR final approve.

5.2 On Approval

- Creates **Staff Obligation Account** (OBL-XX-YY-NNNN)
- disbursement transaction recorded
- Repayment via hr_payroll_line_loans monthly deductions
- Optional cash repayment at Cashier desk

5.3 Exceptional Loans

Above policy limits: **Executive HR → Approvals → MD or Chairman** (exceptional_loan.approve).

6. PAYROLL

6.1 Sign-Off Sequence (MANDATORY)

HR Admin prepares → GM HR approves → MD approves → Lock → Finance pays

Step

Prepare run

GM approve

MD approve

Lock

Export / pay

Critical: Payroll cannot be paid without MD's explicit approval in system.

6.2 Monthly Payroll Procedure

1. **HR → Payroll → Prepare monthly run** for period.
2. Review lines: basic, allowances, deductions, loan recoveries, attendance deductions.
3. Submit for GM HR review.
4. GM HR → **GM approve**.
5. MD → **Executive HR → Compensation → Payroll summary** → MD sign-off.
6. HR → **Lock** run.
7. Finance → **Accounting desk → Payroll** → export bank file + treasury post.
8. Employees view payslips: **My HR → Payslips**.

6.3 PAYE and Pension

- PAYE computed per staff; exported for filing

- Pension: employer + employee portions → GL 2400 (Pension Payable — liability)
-

7. ATTENDANCE

7.1 Daily Roll

- Team HR / HR Admin marks daily attendance
- Permissions: attendance.mark, daily_roll.mark
- Uploads via hr_attendance_uploads for bulk import
- Deductions flow into payroll

7.2 Team HR

Branch managers mark attendance for direct reports at /team-hr/time-absence.

8. DISCIPLINE

8.1 Four-Stage Ladder (Board Resolution)

Stage

- 1
- 2
- 3
- 4

Gross misconduct (theft, fraud, file tampering): may skip to Stage 4.

8.2 Discipline Case Workflow

draft → open → awaiting_employee_response → under_investigation
→ awaiting_hr_review → awaiting_management_decision → action_issued → closed

Case types: query, verbal_warning, written_warning, suspension, gross_misconduct, negligence, absenteeism, theft_fraud, etc.

8.3 Incident Recovery

1. Link material incident to discipline case
 2. Record responsibility weights per party
 3. Generate recovery schedule (hrIncidentRecoveryOps)
 4. Deduct via payroll → staff obligation account
-

9. EMPLOYEE EXIT — 5-STAGE CLEARANCE

Stage

- 1 Initiate

- 2 HR Clearance
- 3 Finance Clearance
- 4 Admin Clearance
- 5 Final Clearance

All 5 stages required before ERP account deactivation.

10. EXECUTIVE HR

10.1 Access

/executive-hr/* — md, ceo, chairman

10.2 Modules

Section

Family & household

Compensation

Approvals

Reports

10.3 Chairman Family Accounts

hr.chairman.manage permission: - School fees for dependents - Family expense payments
- Scholarship payments - Domestic staff management

11. DOCUMENTS AND COMPLIANCE

- Employment letters: hr.letters.generate
 - ID card requests: employee self-service → HR fulfilment
 - Policy acknowledgements: hr_policy_acknowledgements
 - Document expiry alerts on HR Dashboard
 - Staff files retained minimum **7 years**
-

12. TRANSFERS

Transfer Type

Branch transfer

Internal rotation

BM transfer

Enforced in transferTenurePolicy.js.

13. RECRUITING

Public careers: GET/POST /api/public/careers/* (no authentication).

HR manages: hr_job_postings, hr_applicants at /hr/talent → Recruit.

14. HR DASHBOARD KPIs

Active staff · Pending HR review · Awaiting GM final · Payroll awaiting GM · Probation ending · Expiring documents · Open incidents · Queue lines · Overdue SLA

15. SELF-SERVICE COHORTS

Cohort	My HR Sections
Employee	Time off, requests, payslips, loans, documents, ID card, benefits, policies
Scholarship	School, payments, requests (hr_portal_only)
Domestic	Home, payments, documents (executive household staff)

End of SOP-07. Cross-references: SOP-02 (Cashier loan repayment), SOP-03 (Payroll finance), SOP-08 (MD sign-off).

SOP-08: EXECUTIVE OFFICE

Zarewa Aluminium and Plastics Ltd — System of Operations v3.0

Department: Executive Office

System modules: /exec, /executive-hr/*, /manager (when MD acts as BM)

Primary roles: md, ceo, chairman

CEO/Chairman: read-only on transactions; approval authority on sensitive items

1. PURPOSE AND SCOPE

The Executive Office provides strategic oversight, high-value approval authority, and company-wide performance monitoring. The Managing Director operates the business through the **Command Centre** (/exec); CEO and Chairman have read-focused executive views with targeted approval permissions.

2. COMMAND CENTRE

Route: /exec

2.1 Tabs

Tab

Today

Decide

Customers

Trace

Review

Intelligence

Finance

2.2 Period Filters

Key

today

week

month

last_month

custom

2.3 Branch Filters

All · Kaduna · Yola · Maiduguri

3. EXECUTIVE DASHBOARD PANELS

Panel

Sales Summary

Expense Summary

Receivables

Branch Scorecard

BI Analytics

Working Capital

MD Cockpit

Champion Customers

Material Costing (AP3)

Staff Activity

Target vs Actual

MD Attention Inbox
Stock Register Inbox
Payroll Sign-Off Queue
Bank Reconciliation Status
Reserve Policy

4. MD DECISION QUEUE (/exec?tab=decide)

Aggregated pending decisions sorted by tier (critical → urgent → normal) and age:

Decision Type

Credit exceptions (above BM limit)
Below-floor price exceptions
Refunds above ₦1,000,000
Staff loan/leave (exceptional)
Inter-branch loans
Payroll runs
Production gate overrides
Staff purchase credit
Work item decisions
Stock register (stage 3)
Coil short receipt

5. MD APPROVAL AUTHORITY SUMMARY

Area

Refunds
Expenses (payment requests)
Credit exceptions
Price exceptions
Payroll
Inter-branch loans
Receivable write-offs
Delivery override
Treasury reserve policy

6. TREASURY RESERVE POLICY

MD sets minimum cash balance for designated treasury accounts:

1. **Command Centre** → Finance tab → Reserve Policy
 2. Set minimum per account
 3. Accounts below minimum flagged on Executive Dashboard
 4. Branch managers notified for replenishment
-

7. ORG-WIDE GOVERNANCE

7.1 Settings Access (MD)

- Settings → Governance → Office approval thresholds
- Settings → Governance → Org manager targets (sales/production)
- Org policy KV overrides (with audit)
- Price list and pricing policy (/price-list, /pricing-policy)

7.2 Governance Pack

GET /api/reports/governance-pack: - Misaligned refunds - Dual-control warnings - Payment gate breaches - QC gaps

Export for board meetings and audit.

8. CEO AND CHAIRMAN ROLE

8.1 CEO (ceo)

- **Read-only** Command Centre and Reports
- Cannot post transactions, create quotations, or access line-level finance
- Routes: /exec, /reports (if permitted)
- Minimal sidebar: Command Centre → Reports → Account

8.2 Chairman (chairman)

- CEO permissions plus:
 - Executive HR family/scholarship accounts
 - hr.chairman.manage
 - Exceptional loan approval alongside MD
-

9. DAILY MD ROUTINE

Morning (30 minutes)

1. Open **Command Centre** → **Today** tab

2. Review critical alerts (red badges)
3. Open **Decide** tab — clear urgent approvals
4. Check **Payroll sign-off queue** if pay period
5. Review **Reserve policy** warnings

Weekly

1. **Intelligence** tab — coil forecasts, SKU cover
2. **Finance** tab — working capital trend
3. **Reports** → weekly pack
4. All-branch **stock register** progress
5. **Governance pack** review

Monthly

1. Sign stock registers (stage 3) all branches
 2. Review branch scorecard vs targets
 3. Approve month-end payroll
 4. Board pack from Reports → MD operations pack
-

10. BUSINESS INTELLIGENCE

Route: /analytics or /exec?tab=intelligence

- SKU weeks-cover (4-month lookback)
 - Revenue trends and gross margin by branch/product
 - Customer segmentation: champions, at-risk, dormant
 - Cash horizon projections
 - Branch comparative performance
-

11. TRANSACTION TRACING

Trace tab (/exec?tab=trace): - Follow quotation → cutting list → production → delivery → refund → treasury - GET /api/quotations/:id/lifecycle-timeline - Used for customer disputes and audit investigations

12. OFFICIAL NOTICES

MD holds notices.manage: - Company-wide announcements - Require acknowledgement - Pin to board - Set expiry and branch/role targeting

End of SOP-08. Cross-references: All department SOPs for items in Decide queue.

SOP-09: MAINTENANCE

Zarewa Aluminium and Plastics Ltd — System of Operations v3.0

Department: Maintenance

System modules: Machines registry (via Operations/Settings), Maintenance plans and work orders

Primary roles: Maintenance Manager (desig_mm), Maintenance Technician (desig_mtech)

Supporting roles: finance_manager (capex, cost posting), operations_officer

1. PURPOSE AND SCOPE

Maintenance ensures production machinery remains operational, safe, and cost-effective. The ERP tracks machines, meter readings, preventive maintenance plans, corrective work orders, maintenance events, and cost lines linked to GL 5020 (Maintenance COGS).

2. MACHINE REGISTRY

2.1 Machine Records

Field

Machine ID

Name / description

Location

Fixed asset link

Status

2.2 Meter Logs

machine_meter_logs track usage: - Operating hours - Metres produced (where meter fitted) - Fuel consumption links - Used for preventive maintenance scheduling

2.3 Responsibilities

Role	Duties
Maintenance Manager (desig_mm)	Plans, work order approval, technician assignment
Maintenance Technician (desig_mtech)	Execute work orders, log events, parts usage
Production Manager	Report breakdowns; release machine for maintenance window

3. PREVENTIVE MAINTENANCE

3.1 Maintenance Plans (MXPL-YY-NNNN)

Recurring scheduled maintenance:

Plan Type

Daily

Weekly

Monthly

Quarterly

Procedure:

1. Create **Maintenance Plan** linked to machine.
 2. Set frequency, checklist items, estimated duration.
 3. System generates **Work Orders** when due.
 4. MM assigns technician; tracks completion.
-

4. CORRECTIVE MAINTENANCE

4.1 Work Orders (MXWO-YY-NNNN)

Ad-hoc or plan-generated repair jobs:

1. **Report breakdown** — Production or Operations logs issue.
2. MM creates **Maintenance Work Order**:
 - Machine, fault description, priority
 - Link to material request if parts needed
3. Status: Opened → acknowledged → approved → closed
4. Technician executes; logs **Maintenance Event** (MXEV-YY-NNNN).

4.2 Maintenance Events

Completion record: - Date, technician, hours spent - Parts used (linked to stock or expense)
- Labour cost - Root cause notes

4.3 Cost Lines

maintenance_cost_lines classify under: - **Maintenance** expense → GL 5020 (COGS) - May link to **payment request** for external contractor invoices

5. MATERIAL REQUESTS FOR PARTS

When maintenance requires spare parts:

1. Create **Material Request** from work order
 2. Operations fulfils from store or Procurement raises PO
 3. Cost allocated to maintenance work order
-

6. ASSET CUSTODY

Maintenance tools and portable equipment tracked via asset_custody_events:

Event

assign

transfer

confirm_present

report_missing

Linked to HR discipline and recovery schedules if staff negligence established.

7. GATE PASS EVENTS

Equipment leaving site: - Record gate_pass_events with authorisation - Security verification - Return confirmation on re-entry

8. FIXED ASSET LINKAGE

Production machines linked to fixed asset register: - Acquisition cost from capex PO or manual registration - Depreciation via monthly run (60-month useful life for plant) - Disposal through Accounting desk (SOP-03 §8)

9. DOWNTIME REPORTING

On machine breakdown:

1. Production Manager stops job allocation to affected machine
 2. MM creates urgent work order (priority: Critical — 4hr SLA)
 3. Log downtime start/end on work order
 4. Production reschedule communicated to Sales via official notice or memo
 5. Post-incident review if downtime >4 hours
-

10. MAINTENANCE KPIS

KPI

Planned maintenance completion

Breakdown response time

Mean time between failures

Maintenance cost as % of production COGS

Reports: maintenance cost lines in expenses pack; machine meter trends on Operations overview.

11. MONTHLY MAINTENANCE CHECKLIST

- ☐ All due preventive plans completed or rescheduled with reason
 - ☐ Open work orders cleared or escalated
 - ☐ Meter logs entered for all active machines
 - ☐ Asset custody verification for portable tools
 - ☐ Maintenance costs reviewed with Finance (GL 5020)
 - ☐ Critical spares stock level checked with Operations
-

End of SOP-09. Cross-references: SOP-04 (Material requests), SOP-03 (Fixed assets), SOP-07 (Asset custody/discipline).

SOP-10: OFFICE ADMINISTRATION

Zarewa Aluminium and Plastics Ltd — System of Operations v3.0

Department: Office Administration

System modules: / (Workspace), /edit-approvals, Office threads, Forum, Notices

Primary roles: All staff with office.use

Senior roles: md, admin, hr_admin, gmhr (notices.manage)

1. PURPOSE AND SCOPE

Office Administration is the digital workplace for internal communication, structured requests, document filing, and cross-department coordination. Every staff member with office.use accesses the **Workspace** (/) as their collaboration hub.

2. WORKSPACE DESK PROFILES

Desk layout adapts by role (workspaceDeskNav.js):

Profile	Sections
Staff	My Desk, Create Office Record, My Requests, Tasks, Official Notices, Forum, Search
Branch (BM)	Branch Desk, Today's Work, Endorsements, Team Requests, Expense Conversions, Incidents, Branch Forum, Filing, Monitoring, Search
Office (Finance/HR)	Office Desk, Review Queue, Approvals, Expense Conversions, Filing, Notices, Forum, Monitoring, Records, Search
Executive (MD)	Executive Desk, High-value Approvals, Branch Monitoring, Notices, Branch Contributions, Overdue Items, Expense Oversight, Records, Search

3. CREATE OFFICE RECORD WIZARD

3.1 Record Types

Type

Memo
Expense request
Official notice
Forum post
Material request

3.2 Expense Request Procedure

1. **Workspace – Create Office Record – Expense request.**
2. Select **expense category** (see GL mapping in SOP-03).
3. **“Others” category:** minimum 40-character justification required.
4. Add lines: payee, amount, description, supporting attachment.
5. Submit → routes to Branch Manager (≤¥200k) or MD (>¥200k).
6. On approval → appears in **Finance Desk** queue for payout.
7. Filing reference issued: ZR/{branch}/{domain}/{year}/{seq}.

3.3 Smart Memo Composer

Structured internal communications with: - To/From/Cc routing - Subject and body - Attachment support - AI polish optional (/api/office/ai/polish-memo)

4. WORK ITEMS AND SLA

4.1 Unified Inbox

Views: **Action Inbox, Work Tray, Filed, Unfiled**

4.2 SLA by Priority

Priority

Critical

Urgent / High

Normal

4.3 Routing by Role

Role

sales_manager

sales_staff

operations_officer

finance_manager / cashier

md

admin

4.4 Work Item Lifecycle

1. Created (from office record, system event, or manual)
 2. Assigned to desk/role
 3. Decision recorded (work_item_decisions)
 4. Filed with reference
 5. Print snapshot at closure (work_item_print_snapshots)
-

5. EDIT APPROVALS

5.1 When Required

Sensitive edits to locked records: - Confirmed quotations - Received POs - Cleared customer records - Confirmed deliveries

5.2 Procedure

1. User attempts edit → system blocks with **Edit Approval Required**.
2. User submits edit approval request with description of change.
3. Approver (BM or MD) opens /edit-approvals.
4. Reviews change; issues **6-digit one-time code** to requester.
5. Requester enters code → **single save consumed**.

6. Logged in approval_actions and audit_log.

Security: Code expires; cannot reuse; approver cannot be requester.

6. OFFICIAL NOTICES

6.1 Authority

notices.manage: md, admin, hr_admin, gmhr, ceo, chairman

6.2 Publishing Procedure

1. **Workspace → Official Notices → New Notice.**
2. Set title, body, attachments.
3. Target: all staff, specific branch(es), role(s), or department(s).
4. Options:
 - **Require acknowledgement** — staff must click Acknowledge
 - **Pin to board** — stays at top
 - **Expiry date** — auto-archive
5. Publish → official_notices + read/acknowledgement tracking.

6.3 Staff Procedure

1. Check **Official Notices** on Workspace daily.
 2. Read and **Acknowledge** where required.
 3. Unacknowledged notices appear in dashboard alerts.
-

7. COMPANY FORUM

Scope

Branch

Company

Rules: - Professional conduct — discipline policy applies - No customer pricing or confidential financial data - HR may remove posts; gross misconduct triggers discipline case

8. OFFICE DOSSIERS AND FILING

8.1 Dossiers

Document cabinet for filed office records: - office_dossiers + office_dossier_links -
Version history in office_record_versions

8.2 Filing Completeness

Before work item closure: - filingCompleteness.js checks required fields - Documents without filing number cannot be archived - AI-assisted filing: /api/office/threads/:id/ai-file

8.3 Inter-Branch Office Requests

Cross-branch coordination: - Branch manager creates request - Tracked as work item - May convert to material request or payment request

9. CONFIDENTIAL ACCESS

Work items and threads may be marked **confidential**: - Only participants (userIsWorkItemParticipant) or elevated access (workspaceConfidentialAccess.js) can view - Used for sensitive HR, legal, and financial matters - Do not screenshot or forward outside system

10. WORKSPACE SEARCH

GET /api/workspace/search: - Authenticated; results filtered by entity permissions - Search customers, quotations, coils, work items, staff - CEO has empty search (no line-level access)

11. EXPENSE CATEGORY COACHING

System monitors “**Others**” expense category: - >¥50,000: Finance review flag - >15% of branch expenses: coaching alert to BM - MD sees trend on Command Centre

Staff should use specific categories where possible; reserve Others for genuine exceptions.

12. DAILY OFFICE ADMINISTRATION CHECKLIST

All staff: - [] Check Action Inbox on Workspace - [] Read unacknowledged official notices - [] File completed work items (move from Unfiled to Filed)

Branch Manager: - [] Clear endorsement queue - [] Approve expense requests within SLA - [] Review branch forum for issues

MD / Admin: - [] Review executive desk overdue items - [] Publish company notices as needed

13. AI ASSISTANT (ZARE) IN OFFICE CONTEXT

- **Tour this page** on Workspace for navigation coaching
 - Memo polish for formal communications
 - Cannot approve, file, or post on user's behalf
 - Queries logged for knowledge base improvement
-

End of SOP-10. Cross-references: SOP-02 (Expense payout), SOP-08 (MD approvals), SOP-01 (Sales handoffs).

APPENDIX A: GLOSSARY, REFERENCE TABLES & REPORT INDEX

Zarewa Aluminium and Plastics Ltd — System of Operations v3.0

A.1 COMPLETE GLOSSARY

Term	Definition
AP1c	Accounting Phase 1c — production-completion revenue recognition with automatic GL posting
AP2	Accounting Phase 2 — received-basis accounts payable; inventory valued at landed cost
AP3	Accounting Phase 3 — branch P&L, material cost classification, payroll labour per production job
Advance (Customer)	Payment received before quotation; ledger type ADVANCE_IN
Bootstrap	Initial workspace data snapshot loaded on login via GET /api/bootstrap
BM	Branch Manager (role key: sales_manager)
CL	Cutting List — production work order issued from sales
COGS	Cost of Goods Sold — GL accounts 5010–5050
Credit Exception	Formal approval allowing customer credit terms with expiry
CREV	Coil Control Event — immutable log of coil movements
CSRF	Cross-Site Request Forgery token required on all mutating API requests
Delivery Gate	Payment enforcement before dispatch (off/warn/enforce)
FG	Finished Goods — post-production stock
GL	General Ledger

GMHR	General Manager, Human Resources (role key: gmhr)
GRN	Goods Received Note — store receipt against purchase order
GRNI	Goods Received Not Invoiced — AP liability at GRN under AP2
MD	Managing Director (role key: md)
OBL	Staff Obligation Account — unified ledger for loans, credits, recoveries
PAYE	Pay As You Earn — income tax withholding on payroll
POD	Proof of Delivery — customer sign-off on delivery note
PO	Purchase Order
PPGI	Pre-Painted Galvanised Iron — aluzinc coil
QC / QCO	Quality Control / Quality Control Officer
RAG	Retrieval-Augmented Generation — AI knowledge architecture for Zare
RBAC	Role-Based Access Control — 100+ permission keys
RCP	Confirmed Sales Receipt document ID prefix
Reserve Policy	MD-set minimum cash balance for treasury accounts
SLA	Service Level Agreement — work item response windows
SOFP	Statement of Financial Position (balance sheet)
SOP	Standard Operating Procedure
WIP	Work-in-Progress — goods currently in production
Zare	Zarewa built-in AI assistant for navigation and coaching

A.2 COMPLETE GL CHART OF ACCOUNTS

Code	Account Name	Type	Typical Use
1000	Cash / Bank	Asset	Treasury accounts
1200	Staff Loans Receivable	Asset	Staff loan disbursements
1300	Inventory / Closing Stock	Asset	Coil, FG, accessories
1500	Plant & Machinery	Fixed Asset	Production equipment
1501	Land & Buildings	Fixed Asset	Premises
1502	Furniture & Fittings	Fixed Asset	Office furniture
1504	Generator	Fixed Asset	Power equipment
2000	Accounts Payable — Suppliers	Liability	GRN-based payables
2400	Pension Payable	Liability	Payroll pension contributions
2500	Customer Advances / Refunds	Liability	Unapplied advances,

	Payable		approved refunds
3200	Chairman Withdrawal	Equity	Equity draw — not expense
4000	Sales Revenue	Income	AP1c production recognition
5010	Fuel & Lubricant	COGS	Production fuel
5020	Maintenance	COGS	Plant maintenance
5030	Outside Corrugation	COGS	Sub-contracted corrugation
5040	Production Cost / Accessories	COGS	Direct production costs
5050	Purchases / Carriage Inward	COGS	Raw materials, freight
6000	Wages	COGS/OpEx	Direct wages
6100	Depreciation	OpEx	Non-cash depreciation
6110	Admin Salary	OpEx	Office salaries
6120	Admin / Office / HQ / Others	OpEx	General overhead
6130	Rent & Utilities	OpEx	Premises costs
6140	IT & Software	OpEx	Technology
6150	Insurance	OpEx	Premiums
6160	Professional Fees / Tax	OpEx	Audit, legal, levies
6170	Bank Charges	OpEx	Bank fees
6180	Zakat & Sallah	OpEx	Religious obligations
6300	Interest Expense	OpEx	Finance charges
6400	Marketing & Advertising	OpEx	Promotional spend

A.3 HR PERMISSION BUNDLES

Bundle	Roles	Key Permissions
selfService	All staff	hr.self, my_profile.view, my_leave.request, my_loan.request, my_payslip.view
branchManager	sales_manager	+ hr.team.view, attendance.mark, leave.endorse, loan.endorse
hrAdmin	hr_admin	staff.manage, all exit stages, asset custody, recovery.manage
gmhr	gmhr	gm_approve, final_approve, exit.final_clear, letters.approve
mdExecutive	md	executive.view, payroll.md_approve, exceptional_loan.approve
executiveScholarshipDomestic	ceo, chairman	chairman.manage, benefits, payroll.view_sensitive
financeHr	finance_manager	payroll.pay, payroll.export, loan.disburse,

A.4 REFUND REASON CATEGORIES

Category

Order cancellation
Unproduced meterage
Overpayment
Transport issue
Installation issue
Additional services
Accessory shortfall
Stone flatsheet shortfall
Calculation error
Substitution difference
Customer commission
Other

A.5 REPORTS INDEX

Report

Summary counts
Pending approvals
Production status
Governance pack
Receipts register
AR as-at
Sales bridge
Expenses pack
Refunds pack
Purchase register
Stock coil as-at
MD operations pack
Daily pack
Weekly pack
Material transaction
Revenue production
Business intelligence

A.6 APPLICATION ROUTE MAP

Route

/

/sales

/customers

/customers/:id

/procurement

/operations

/operations/coils/:coilNo

/operations/material-exceptions

/accounts

/accounting

/manager

/exec

/reports

/analytics

/hr/*

/team-hr/*

/executive-hr/*

/my-profile/*

/settings/*

/edit-approvals

/price-list

/pricing-policy

A.7 ENVIRONMENT VARIABLES (OPERATIONS)

Variable

SESSION_TIMEOUT_MINUTES

FAILED_LOGIN_LOCK_THRESHOLD

ACCOUNT_LOCK_MINUTES

DELIVERY_PAYMENT_GATE

ENFORCE_DUAL_CONTROL_PAYMENTS

REFUND_MD_APPROVAL_THRESHOLD_NGN

EXPENSE_MD_APPROVAL_THRESHOLD_NGN

ZAREWA_LEDGER_POST_MAX

A.8 DESIGNATION CODES

Code

desig_md
desig_edo
desig_edf
desig_edc
desig_gmhr
desig_bm
desig_po
desig_hop
desig_pm
desig_qco
desig_mm
desig_mtech
desig_hrrep
desig_aud

End of Appendix A. Return to [00-MASTER-INDEX.md](#) for document navigation.

ANNEX B: DETAILED SCENARIO WALKTHROUGHS

Zarewa Aluminium and Plastics Ltd — System of Operations v3.0

Purpose: End-to-end worked examples mapping real business scenarios to exact system steps.

SCENARIO B1: STANDARD QUOTE-TO-CASH-TO-DELIVERY (KADUNA BRANCH)

Situation: Customer Alhaji Musa Ibrahim wants 250 metres of 0.45 mm aluzinc long-span roofing in Charcoal Grey for his building project in Kaduna. Total quotation value: ₦2,875,000. Customer pays 80% advance by bank transfer.

Actors

- Sales Officer: Amina (sales_staff)

- Cashier: Ibrahim (cashier)
- Branch Manager: Yusuf (sales_manager)
- Operations Officer: Hassan (operations_officer)
- Production: factory team Yola (operations_officer at Yola branch)

Step-by-Step

Day 1 — Customer and Quotation (Sales Office)

1. Amina signs in; confirms workspace branch = Kaduna (BR-KD).
2. Navigates to /sales → Customers → searches phone 08031234567.
3. No match found → creates new customer:
 - Name: Alhaji Musa Ibrahim
 - Phone: 08031234567
 - Address: Plot 42, Barnawa, Kaduna
 - Branch: BR-KD (auto)
4. Creates quotation QT-KD-26-0156:
 - Line 1: Long-span aluzinc, 0.45 mm, Charcoal Grey, 250 m @ ₦11,500/m = ₦2,875,000
 - Payment terms: Partial advance
 - Validity: 30 days
5. System validates price against floor — at floor; no exception needed.
6. Saves quotation; status = active.

Day 1 — Payment (Cashier Desk)

7. Customer transfers ₦2,300,000 (80%) to Zarewa GTBank account.
8. Amina (or Ibrahim) opens quotation → Record Payment:
 - Amount: ₦2,300,000
 - Date: today
 - Treasury account: GTBank Kaduna Main
 - Reference: “TRF Musa Ibrahim 250m roofing 80pct”
9. Amount > ₦100,000 → types amount twice.
10. Posts receipt → RCP-KD-26-0091 created; status Pending clearance.
11. Ibrahim opens /accounts → Receipts → locates RCP-KD-26-0091.
12. Verifies bank alert matches → Confirm Payment Received.
13. Status = Cleared. Quotation payment_status = Partial (80%).

Day 2 — Cutting List (Sales Office)

14. Amina opens QT-KD-26-0156 → Create Cutting List.
15. Payment gate: 80% > 70% threshold — gate passes.
16. Enters cutting list CL-KD-26-0042:
 - Line 1: 250 m long-span 0.45 Charcoal Grey

17. Saves; status cutting_list_issued.

Day 2 — Production Release (Operations — Yola)

18. Hassan at Yola switches workspace to Yola (BR-YOL) or receives inter-branch cutting list per branch policy.
19. /operations → Production line → sees CL-KD-26-0042 in queue.
20. Releases to production → PJ-YL-26-0088 created (Planned).
21. Allocates coil lot CL-2024-8847 (0.45 mm Charcoal Grey, 4,200 kg remaining).
22. Starts job → Running. WIP opened.

Day 3 — Production Complete (Yola)

23. Factory produces 248 m actual (2 m variance — within 5%).
24. Hassan completes job:
 - Produced metres: 248 m
 - Conversion band: OK
25. AP1c auto-posts:
 - Dr AR ₦2,852,000 (248m × ₦11,500)
 - Cr Revenue ₦2,852,000
26. Job status = Completed.

Day 4 — Delivery (Operations — Kaduna)

27. 248 m transported Kaduna → Hassan (Kaduna) creates DLV-KD-26-0033.
28. Delivery gate: 80% paid — warn mode shows notice; proceeds.
29. Customer signs delivery note.
30. Hassan confirms delivery with POD reference.
31. Stock deducted; quotation status progresses toward completed.

Day 5 — Balance Payment

32. Customer pays balance ₦552,000 (remaining on adjusted 248 m + any agreed adjustment).
33. Receipt posted and cleared.
34. Quotation payment_status = Paid.

System Records Created

QT-KD-26-0156 → CL-KD-26-0042 → PJ-YL-26-0088 → DLV-KD-26-0033 → RCP-KD-26-0091, RCP-KD-26-0098 → GL entries via AP1c

SCENARIO B2: BELOW-FLOOR PRICING WITH MD EXCEPTION

Situation: Competitor undercuts market. Customer wants 500 m at ₦10,800/m (floor is ₦11,200/m). BM agrees commercially; MD must approve before production.

1. Sales creates quotation with unit price ₦10,800/m.
2. System shows **Below Floor** warning on save — allowed.
3. Cutting list blocked — error message references price exception.
4. BM notifies MD with competitor quote evidence.
5. MD opens quotation → Price Exception panel → approves with note: “Competitor XYZ quote dated 20/06/2026 attached. One-time match to secure 500m order.”
6. md_price_exception_approved_at_iso recorded.
7. Cutting list now creatable.
8. Production proceeds normally.

Audit trail: quotation record, approval_actions, MD user ID.

SCENARIO B3: CUSTOMER REFUND — UNPRODUCED METERAGE

Situation: Customer paid 100% on 300 m order. Production completed only 180 m due to coil defect. Customer requests refund for 120 m unproduced.

1. Sales Officer opens QT → Refunds → New.
2. Category: **Unproduced meterage**.
3. System preview suggests 120 m × unit price = refund amount.
4. Production alignment check shows 180 m completed — consistent.
5. Submits refund → Pending.
6. BM opens /manager → Cash out → reviews:
 - Lifecycle timeline: quote → CL → PJ (180m) → delivery
 - Payment 100% cleared
 - Approves ₦1,380,000 (120 × ₦11,500).
7. Ibrahim (Cashier) pays from Desk queue.
8. Treasury REFUND_PAYOUT movement created.
9. Customer receives bank transfer.

If amount were ₦1,500,000 (>₦1M threshold): MD must approve, not BM alone.

SCENARIO B4: COIL PROCUREMENT TO GRN (FULL CHAIN)

Situation: MD orders 20 tonnes aluzinc coil 0.50 mm from Supplier SteelCo.

1. MD creates PO-KD-26-0024 at /procurement:
 - Supplier: SteelCo Nigeria Ltd
 - Line: Coil 0.50 mm, 20,000 kg @ ₦485/kg
 - Transport: Haulier Express, ₦180,000 haulage
2. Approves PO → Approved.
3. Assigns transport → On loading.

4. Pays haulage → In Transit.
5. Coil arrives Yola — Operations posts GRN:
 - Coil nos: SC-2026-4410, SC-2026-4411 (two rolls)
 - Weights: 10,200 kg + 9,850 kg = 20,050 kg (50 kg over PO — within tolerance)
6. AP2: Dr Inventory, Cr AP at landed cost.
7. PO status → Received.
8. Finance pays SteelCo per payment terms.

Short receipt variant: If received 18,500 kg vs 20,000 ordered (1,500 kg short = 7.5%): - Above tolerance → MD work item auto-created. - MD decides: accept partial, claim supplier, or cancel balance.

SCENARIO B5: MONTHLY PAYROLL (FULL CHAIN)

Situation: June 2026 payroll for 47 Kaduna staff.

1. HR Admin (Fatima) prepares payroll run PR-2026-06-KD.
2. Reviews lines: basic, transport allowance, loan deductions (3 staff), attendance deductions (2 staff).
3. Submits for GM HR review.
4. GM HR (Grace) → GM approve.
5. MD opens Executive HR → Compensation → Payroll summary → MD sign-off.
6. Fatima locks payroll run.
7. Finance Manager exports bank file from Accounting desk.
8. Executes bulk transfer June 28.
9. Posts treasury movements.
10. Staff view payslips on My HR June 29.

Blocked scenario: If MD forgets sign-off → lock button disabled with message referencing md_approved_at_iso.

SCENARIO B6: MATERIAL INCIDENT — COIL DAMAGE

Situation: Forklift damages coil CL-2024-7721 in yard. 340 kg declared unusable.

1. Hassan logs material incident MINT-KD-26-0007:
 - Type: coil_damage
 - Coil: CL-2024-7721
 - Before: 2,100 kg → After: 1,760 kg
 - Storekeeper + operator names recorded
2. Saves draft → photographs attached → Submits.
3. BM Yusuf reviews → Approve & post.

4. Coil qty_remaining reduced 340 kg.
 5. If negligence established → HR discipline case linked.
 6. Recovery schedule if staff fault → payroll deduction via OBL account.
-

SCENARIO B7: EXPENSE REQUEST — BRANCH MOTOR FUEL

Situation: Sales team needs ₦85,000 diesel for generator during power outage.

1. Sales Officer creates Office Record → Expense request:
 - Category: Fuel & lubricant (GL 5010)
 - Payee: Total Energies station
 - Amount: ₦85,000
 - Justification: “Generator fuel 3 days power outage week 24 — sales office operational continuity”
 2. Routes to BM Yusuf (below ₦200k threshold).
 3. Yusuf approves on Manager dashboard.
 4. Ibrahim pays from Finance Desk.
 5. Filing reference ZR/KD/expense/2026/0142 issued.
-

SCENARIO B8: LEAVE REQUEST — FULL WORKFLOW

Situation: Employee John (L3, 14 days annual leave) requests 5 days annual leave.

1. John → My HR → Time off → Request:
 - Type: Annual
 - Dates: 10–14 July 2026
 2. Status: submitted → hr_review.
 3. HR Admin triages → hr_review approved.
 4. John's line manager (BM) → Team HR → Endorses.
 5. GM HR → final approval.
 6. Leave balance: $14 - 5 = 9$ days remaining.
 7. Calendar updated; payroll notified if unpaid leave (not applicable here).
-

SCENARIO B9: EDIT APPROVAL ON LOCKED QUOTATION

Situation: Customer changes colour after quotation confirmed but before production.

1. Sales attempts edit on QT-KD-26-0200 → blocked.
2. Submits edit approval request: “Customer changed Charcoal Grey to Ocean Blue — same price.”
3. BM opens /edit-approvals → reviews → issues code 847291.
4. Sales enters code → changes colour field → saves once.

5. Code consumed; further edits require new approval.
-

SCENARIO B10: INTER-BRANCH LOAN

Situation: Yola factory cash short; Kaduna HQ lends ₦5,000,000.

1. Finance Manager proposes inter-branch loan:
 - Lender: BR-KD
 - Borrower: BR-YOL
 - Amount: ₦5,000,000
 - Purpose: Supplier payment bridge
 2. MD approves on Command Centre → Decide tab.
 3. Treasury movements executed both branches.
 4. Loan status: executed.
 5. Yola repays ₦2,500,000 month 1, ₦2,500,000 month 2.
 6. Loan status: settled.
-

End of Annex B — Scenario Walkthroughs

ANNEX C: IT OPERATIONS, SECURITY & DISASTER RECOVERY

Zarewa Aluminium and Plastics Ltd — System of Operations v3.0

Audience: IT administrators, MD (oversight), external IT contractors

C.1 IT GOVERNANCE

C.1.1 Responsibilities

Role	Responsibility
System Administrator (admin)	User management, deployments, backups, migrations
MD	Approve production changes, credential policy
Finance Manager	Period lock policy, integration API keys approval
All staff	Report incidents, protect credentials

C.1.2 Change Management

1. All production deployments follow change window (typically Sunday 02:00–06:00 WAT).
2. Run `npm run test` and `npm run test:e2e` before deploy.

3. Database migrations auto-run on startup — review server/migrate.js changelog before deploy.
 4. Rollback plan: restore database backup + previous API build.
-

C.2 REPOSITORY STRUCTURE

Repository

Zarewa-backend-main

Zarewa-frontend-main

Key backend paths

- server/index.js — API entry
- server/httpApi.js — route registration
- server/auth.js — roles and permissions
- server/schemaSql.js — database DDL
- server/migrate.js — incremental migrations
- shared/ — cross-cutting business rules

Key frontend paths

- src/App.jsx — routing
 - src/lib/moduleAccess.js — RBAC UI
 - src/lib/roleTrainingGuide.js — onboarding
 - src/lib/helpOperationalCatalog.js — Zare knowledge (~ 1000 entries)
-

C.3 ENVIRONMENT CONFIGURATION

C.3.1 Required Production Variables

Variable

NODE_ENV

ZAREWA_DB_URL or MySQL vars

SESSION_SECRET

COOKIE_SECURE

CORS_ORIGIN

ZAREWA_STATIC_DIR

C.3.2 Security Variables

Variable

ENFORCE_DUAL_CONTROL_PAYMENTS

DELIVERY_PAYMENT_GATE

SESSION_TIMEOUT_MINUTES

FAILED_LOGIN_LOCK_THRESHOLD

C.3.3 Optional Features

Variable

ZAREWA_AI_API_KEY

ZAREWA_EMPTY_SEED

VITE_API_BASE

See docs/ENVIRONMENT.md and docs/DEPLOYMENT.md in backend repo.

C.4 DEPLOYMENT PROCEDURES

C.4.1 Combined Deployment

1. Build frontend: cd Zarewa-frontend-main && npm run build
2. Set ZAREWA_STATIC_DIR to dist/ path
3. Start API: cd Zarewa-backend-main && npm start
4. Verify: GET /api/health returns 200
5. Verify: login page loads, bootstrap succeeds

C.4.2 Split Deployment

1. Deploy API to VM with HTTPS
2. Deploy frontend static files to CDN/host
3. Set VITE_API_BASE to API URL at frontend build time
4. Configure CORS_ORIGIN on API
5. Cookie SameSite=None; Secure=true for cross-origin

C.4.3 Health Checks

Endpoint

GET /api/health

GET /api/readyz

GET /api/livez

Monitor these every 60 seconds in production.

C.5 DATABASE ADMINISTRATION

C.5.1 Backup Schedule

Frequency

Daily

Weekly

Pre-deploy

C.5.2 Backup Procedure (MySQL)

```
mysqldump -u zarewa -p zarewa_prod > zarewa_backup_$(date +%Y%m%d).sql
```

Verify backup integrity weekly by restore to staging environment.

C.5.3 Migration Safety

1. Migrations run in transactions where supported
2. Tracked in zarewa_schema_migrations table
3. Never edit applied migrations — create new migration file
4. Test migrations on staging copy of production data first

C.5.4 Data Reset (NON-PRODUCTION ONLY)

Admin data reset available in Settings for development/training environments. **Never run data reset in production without MD written authorisation.**

C.6 SECURITY HARDENING CHECKLIST

Pre-Production (Mandatory)

- ☐ Replace all demo passwords
- ☐ Remove or disable demo viewer account
- ☐ HTTPS enabled with valid certificate
- ☐ COOKIE_SECURE=true
- ☐ ENFORCE_DUAL_CONTROL_PAYMENTS=1
- ☐ Strong SESSION_SECRET (32+ random bytes)
- ☐ Database credentials not in source code
- ☐ API not exposed to public internet without firewall
- ☐ Admin account limited to named individuals (not shared)
- ☐ Integration API keys rotated from defaults

Ongoing

- ☐ Quarterly credential rotation (admin, integration keys)
 - ☐ Review customPermissionAudit monthly
 - ☐ Review failed login attempts weekly
 - ☐ Patch Node.js dependencies monthly (npm audit)
 - ☐ Verify backup restore quarterly
-

C.7 INCIDENT RESPONSE

C.7.1 Severity Levels

Level

P1 Critical

P2 High

P3 Medium

P4 Low

C.7.2 P1 Response Procedure

1. Confirm outage via /api/health and user reports
2. Notify MD and department heads
3. Check server process, database connectivity, disk space
4. Review recent deployments — rollback if correlated
5. Communicate ETA to staff via official notice
6. Post-incident review within 48 hours

C.7.3 Data Integrity Incident

If suspected data corruption or unauthorised access:

1. Isolate affected system (read-only mode if available)
 2. Preserve audit logs — do not delete
 3. Notify MD and external auditor if required
 4. Restore from last known-good backup to staging
 5. Compare affected records; determine scope
 6. Correct through controlled reversals (not direct DB edits)
 7. Document in incident report
-

C.8 DISASTER RECOVERY

C.8.1 Recovery Time Objectives

Scenario

API server failure

Database corruption

Complete site loss

C.8.2 DR Procedure

1. Provision replacement server
2. Restore latest database backup
3. Deploy latest verified API + frontend builds
4. Update DNS to new server

5. Verify health checks and login
6. Run reconciliation: treasury balances, open quotations, production queue
7. Communicate restoration to all branches

C.8.3 Degraded Mode Operations

If ERP unavailable >4 hours during business hours:

1. BM authorises paper-only recording for receipts and GRNs
 2. **Single designated person** maintains paper log
 3. Double-entry prohibited on paper — one writer
 4. Full catch-up within 24 hours of restoration
 5. IT supervises catch-up to prevent duplicates
-

C.9 TESTING AND QA

C.9.1 Automated Test Suites

Suite

Unit tests

E2E tests

Key E2E specs: - sales-gate.spec.js — payment and price gates - sales-refund-finance-checklist.spec.js — refund dual control - access-control.spec.js — role restrictions - hr-smoke.spec.js — HR workflows - operational-sop-matrix-500.spec.js — SOP route coverage

C.9.2 Pre-Release Checklist

- ☐ All unit tests pass
 - ☐ E2E smoke pass on staging
 - ☐ Migration tested on staging DB copy
 - ☐ RBAC regression check (cashier cannot approve refund)
 - ☐ MD payroll gate still enforced
 - ☐ Backup taken immediately before deploy
-

C.10 INTEGRATION API

C.10.1 Read-Only GL Export

External accounting or BI tools may consume:

- GET /api/integration/v1/trial-balance — Bearer token auth
- GET /api/integration/v1/journals — journal register export

C.10.2 Key Management

1. Create key in Settings → Integrations
 2. Assign descriptive name and expiry
 3. Distribute securely to integration owner
 4. Rotate quarterly; revoke on staff departure
 5. Monitor usage in audit log
-

C.11 MONITORING AND LOGS

C.11.1 Application Logs

- API logs: stdout/stderr captured by process manager (pm2, systemd)
- Audit log: audit_log table — query via Settings or SQL
- HR audit: hr_audit_events table

C.11.2 Workspace Monitoring

/workspace/monitoring — admin/exec view of: - Active sessions - Office activity levels - Work item SLA breaches

C.12 USER SUPPORT TIERS

Tier

L1

L2

L3

Zare is L1 support — staff should try Zare “Tour this page” before escalating.

End of Annex C — IT Operations

ANNEX D: COMPLIANCE, CONTROLS & AUDIT FRAMEWORK

Zarewa Aluminium and Plastics Ltd — System of Operations v3.0

Audience: Internal audit, finance, MD, external auditors

D.1 INTERNAL CONTROL FRAMEWORK

Zarewa ERP implements a three-layer control model:

Layer	Mechanism	Examples
Preventive	Blocks invalid action before it occurs	Period lock, duplicate receipt block, refund self-approval block
Detective	Alerts and reports after event	Governance pack, exception queue, dual-control warnings
Corrective	Reversal and adjustment workflows	Receipt reversal, material incident void, edit approval

D.2 SEGREGATION OF DUTIES MATRIX (COMPLETE)

Process	Initiate	Approve	Execute	Record	Reconcile
Customer sale	Sales	BM (clearance)	Operations (delivery)	System auto	Finance (AR)
Customer receipt	Sales/ Cashier	Cashier (clearance)	Cashier (post)	System auto	Finance (bank rec)
Customer refund	Sales	BM/MD/ Finance	Cashier/ Finance	System auto	Finance
Expense payment	Staff	BM/MD	Cashier/ Finance	System auto	Finance
Purchase order	Procurement/MD	MD	—	System	Finance (AP)
Goods receipt	Operations	—	Operations	System auto	Store (stock register)
Supplier payment	Finance	Finance	Finance	System auto	Finance (AP rec)
Production job	Operations	BM (QC)	Operations	System auto	Finance (AP1c)
Payroll	HR Admin	GMHR + MD	Finance	System auto	Finance
GL journal	Finance	—	Finance	Finance	Finance (TB)
Period close	—	—	Finance	Finance	MD review
User creation	Admin	MD policy	Admin	System auto	Admin audit
Price list change	MD	—	MD	System auto	Sales review
Material incident	Operations	BM	System auto-post	System	Store

D.3 FINANCIAL CONTROLS DETAIL

D.3.1 Revenue Recognition Control (AP1c)

Risk: Revenue recognised before production or after cash only.

Control: AP1c auto-posts at production job completion only.

Test: Select completed jobs; verify GL Cr 4000 matches produced metres × price.

Frequency: Monthly.

D.3.2 Cash Receipt Control

Risk: Fictitious receipts inflate sales.

Control: Cashier clearance required; amounts ≥ ₦100k double-entered.

Test: Sample 25 receipts; trace to bank statement.

Frequency: Weekly.

D.3.3 Refund Control

Risk: Fraudulent refunds to colluding parties.

Control: Requester ≠ approver ≠ payer; MD gate > ₦1M; production alignment check.

Test: All refunds > ₦500k monthly; verify evidence on file.

Frequency: Monthly.

D.3.4 Procurement Control

Risk: Kickbacks, unauthorised purchases.

Control: MD approval on POs; three-way match at month-end.

Test: Sample 10 POs; trace PO → GRN → payment.

Frequency: Monthly.

D.3.5 Inventory Control

Risk: Coil theft, unrecorded usage.

Control: Coil control events immutable; stock register 4-stage sign-off.

Test: Physical coil count vs system for 5 random lots.

Frequency: Monthly (at stock register).

D.4 HR AND PAYROLL CONTROLS

D.4.1 Ghost Employee Risk

Control: Payroll run requires active HR profile; exit clearance blocks deactivation bypass.

Test: Reconcile payroll headcount to HR active staff report.

D.4.2 Unauthorized Salary Change

Control: Salary matrix changes require salary_structure.approve (MD).

Test: Review all salary changes in period against approved requests.

D.4.3 Loan Fraud

Control: Loan approval chain; obligation account tracks balance; payroll deduction auto-calculated.

Test: OBL balances agree to loan agreements on file.

D.5 AUDIT LOG REVIEW PROCEDURES

D.5.1 Daily Review (IT Admin)

Query audit_log for: - login.failed — brute force attempts - permission.override — custom permission changes - admin.data_reset — must be empty in production - refund.dual_control.admin_trial — admin bypass flags

D.5.2 Weekly Review (Finance Manager)

- Receipt reversals
- Refund approvals and payouts
- Period lock attempts
- Treasury corrections

D.5.3 Monthly Review (MD + Internal Audit)

- Governance pack export
- All refunds > ₦1,000,000
- Others expense category > 15% branches
- Dual-control warnings
- Edit approval usage

D.5.4 Audit Log Export

GET /api/audit/export.ndjson — admin only; full export for external auditor.

D.6 REGULATORY COMPLIANCE

D.6.1 PAYE and Tax

- Payroll computes PAYE per staff per Nigerian tax bands
- Export from HR payroll module for FIRS filing
- Finance retains payslips and tax schedules 6 years

D.6.2 Pension (PENCOM)

- Employer and employee contributions calculated per statute
- GL posts to 2400 Pension Payable (liability)
- Monthly remittance schedule exported with payroll

D.6.3 Data Protection

- Staff personal data in hr_staff_profiles — access restricted by HR permissions
 - Bank details encrypted at rest (hrBankCrypto.js)
 - Confidential work items restricted to participants
 - Staff files retained 7 years post-separation; then secure destruction per policy
-

D.7 FRAUD INDICATORS AND RESPONSE

Indicator	Investigation Step
Multiple refunds to same customer in 30 days	Review lifecycle timeline; interview sales officer
Receipts cleared without matching bank credit	Bank rec investigation; cashier interview
Production metres consistently High conversion	QC review; possible metre inflation
Others expenses spike at branch	BM coaching; MD review
Same user login from two IPs simultaneously	Session security review; password reset
Coil qty drops without production job	Material incident search; CCTV if available
Payroll line for separated employee	Exit clearance audit

Response: Suspend ERP access pending investigation (Admin); preserve audit logs; escalate to MD; law enforcement if criminal.

D.8 EXTERNAL AUDIT PREPARATION

Documents to Prepare

Package

Trial balance
GL activity by account
Receipts register
Purchase register
Stock register (locked)
Payroll summary + MD sign-off
Bank reconciliation
Governance pack

Audit log export

Related party transactions

Auditor System Access

Provide viewer role or read-only custom role with reports.view — never admin credentials.

D.9 CONTROL DEFICIENCY REPORTING

Staff who identify control weaknesses:

1. Report to line manager and Finance Manager (if financial)
 2. Do not exploit weakness
 3. IT logs remediation ticket
 4. MD notified if material
 5. Fix verified in next audit cycle
-

D.10 BOARD REPORTING (QUARTERLY)

MD presents to Board from Command Centre:

- Branch scorecard vs targets
 - Governance pack summary
 - Material incidents and discipline summary
 - Stock register status all branches
 - AP1c/AP2/AP3 readiness
 - IT incident summary
 - Fraud indicators (if any)
-

End of Annex D — Compliance & Audit Framework

ANNEX E: EXTENDED DEPARTMENT PROCEDURES & TROUBLESHOOTING

Zarewa Aluminium and Plastics Ltd — System of Operations v3.0

E.1 SALES OFFICE — EXTENDED PROCEDURES

E.1.1 Quotation Line Types Reference

Category

Long-span aluzinc
Step-tile aluzinc
Stone-coated
Ridge / flashing
Gutter / accessory
Transport
Installation

E.1.2 Quotation Validity Management

1. Default validity: 30 days from quotation date (configurable per quotation).
2. System generates follow-up alert at 7 days before expiry.
3. Sales officer actions before expiry:
 - Contact customer (log CRM interaction)
 - Extend validity (edit quotation if not locked)
 - Archive if customer declined
4. Expired quotations cannot release cutting list without BM reactivation.

E.1.3 Multi-Branch Sales Scenarios

Customer in Kaduna, production at Yola: 1. Quotation created at Kaduna (BR-KD) — branch-scoped. 2. Cutting list issued at Kaduna. 3. Yola operations sees cutting list if shared via org policy or inter-branch workflow. 4. Production job created at Yola (BR-YOL). 5. Delivery may be from Kaduna stock or in-transit from Yola.

Branch bar: Always confirm which branch owns the customer and quotation before posting.

E.1.4 Sales Troubleshooting

Error Message	Cause	Resolution
“Cutting list blocked — payment”	Paid % below branch threshold	Post more receipts or BM production override
“Cutting list blocked — price exception”	Below-floor price without MD approval	MD price exception approval
“Duplicate customer phone”	Phone exists on another customer	Search existing; do not create duplicate
“Quotation locked”	Edit approval	Request 6-digit code from BM

“Refund blocked — uncleared receipts”	required Receipts pending clearance	Cashier must clear first
“Order cancellation blocked”	Production completed	Use Unproduced meterage or Other with evidence

E.2 CASHIER DESK — EXTENDED PROCEDURES

E.2.1 Treasury Account Selection Guide

Payment Method

Cash at office

Bank transfer

POS card payment

Cheque

Wrong account selection: Reverse receipt; re-post to correct account. Do not leave misclassified.

E.2.2 Partial Receipt Scenarios

Customer pays in instalments: 1. Each instalment = separate receipt line on same quotation. 2. Never combine two physical payments into one receipt with wrong date. 3. Payment status updates cumulatively: Partial until ≥99.5% paid.

Customer overpays: 1. Post full amount received. 2. System may create overpayment balance. 3. Options: apply to next quotation, refund overpayment, or hold as advance.

E.2.3 End-of-Day Reconciliation Form (Paper Backup)

Field

Date

Cashier name

Opening cash balance (system)

Opening cash balance (physical)

Total receipts confirmed today

Total payouts today

Closing cash balance (system)

Closing cash balance (physical)

Variance

Signed

Retain paper form 90 days even when system reconciled.

E.2.4 Cashier Troubleshooting

Issue	Resolution
Cannot confirm receipt	Verify cashier.receipts.confirm permission
Refund pay button disabled	Check approver ≠ you; verify Approved status
Treasury balance negative	Escalate Finance Manager — data or sequencing error
Duplicate receipt error	Verify not already posted; use override only with BM approval
Period closed error	Finance must open period or use current date

E.3 ACCOUNTING DESK — EXTENDED PROCEDURES

E.3.1 Manual Journal Entry

When required (corrections, accruals, reclassifications):

1. Accounting desk → GL → Manual journal.
2. Enter period key (YYYY-MM) — must be open.
3. Add lines: account code, debit/credit, description, reference.
4. **Debits must equal credits.**
5. Post → journal ID assigned.
6. Document supporting memo in office dossier.

Restrictions: No manual journals to Revenue (4000) for customer sales — use AP1c production path.

E.3.2 Opening Pack (New Branch / Cutover)

For new branch go-live or accounting cutover:

1. Accounting → Opening Pack.
2. Register roll-up: enter opening balances per GL account.
3. Capital entry if applicable.
4. One balanced journal post to establish opening position.
5. Verify trial balance before first operational posting.

E.3.3 AP1c Dry-Run Procedure

Before enabling AP1c in production:

1. Accounting → AP1c → Dry run.
2. Select date range with completed production jobs.
3. Review proposed GL entries without posting.
4. Compare to expected revenue from production report.

5. Resolve discrepancies with IT if systematic.
6. MD and Finance Manager sign cutover memo.
7. Enable AP1c posting.

E.3.4 Month-End Journal Entries (Standard)

Entry

Depreciation
Closing stock
Payroll accrual
Pension accrual
Bank charges
,

E.4 OPERATIONS — EXTENDED PROCEDURES

E.4.1 Coil Specification Matching

Before allocating coil to job, verify:

Quotation Spec

Gauge 0.45 mm
Charcoal Grey
Long-span profile
Aluzinc

Substitution: If coil colour/gauge substituted, sales must document on quotation; may affect refund if price difference.

E.4.2 Stone-Coated Production

1. Stone lines tracked in sheets; system converts via stoneFlatsheetSheetsToM2.
2. production_job_stone_flatsheet_usage records consumption.
3. Stone inventory separate from coil register.
4. GRN stone materials via Procurement stone PO.

E.4.3 Accessory Fulfillment

1. production_job_accessory_usage links accessories to job.
2. Partial fulfillment across multiple deliveries supported.
3. Shortfall may trigger refund category Accessory shortfall.

E.4.4 Yard Coil vs Store Coil

- **Yard coils:** yard_coils — outdoor storage register
- **Store coils:** coil_lots — production-ready register
- Transfer yard → store via coil control event before production allocation.

E.5 PRODUCTION — EXTENDED PROCEDURES

E.5.1 Conversion Variance Deep Dive

Theoretical metres = coil kg consumed ÷ standard kg/m (from conversion table per gauge).

Variance %

0–3%

3–5%

5–10%

>10%

E.5.2 Multi-Coil Jobs

Large orders may span multiple coils:

1. Allocate coil 1 → partial production → return unused kg.
2. Allocate coil 2 → continue until total metres reached.
3. Each allocation logged in production_job_coils.
4. Total consumption = sum across coils.

E.5.3 Job Cancellation

If production cannot complete (coil defect, machine breakdown):

1. Cancel job with reason (requires manager).
 2. Release coil reservations.
 3. WIP reversed per AP3 rules.
 4. Sales notified for customer communication.
 5. May trigger refund or re-scheduling.
-

E.6 PROCUREMENT — EXTENDED PROCEDURES

E.6.1 Supplier Evaluation Criteria

Before adding new supplier to master:

Criterion

Company registration

Tax compliance

Quality certification

Bank verification

Reference customers

Price competitiveness

MD approves new strategic suppliers.

E.6.2 Coil PO Specification Template

Every coil PO line must specify: - Gauge (mm) - Colour RAL or trade name - Profile/design - Total kg - Price per kg (¥) - Origin/mill if required - Expected coil count (rolls) - Tolerance (+/- % on weight)

E.6.3 Transport Agent Management

1. Register agent with bank details and coverage routes.
 2. Link to PO transport field.
 3. Haulage cost part of landed cost (AP2).
 4. Transport catch-up tab reconciles in-transit vs agent invoices.
-

E.7 HR — EXTENDED PROCEDURES

E.7.1 Probation Review

At probation end (default 6 months):

1. HR Dashboard alerts approaching probation end.
2. Line manager submits performance assessment.
3. HR Admin records outcome:
 - Confirm permanent employment
 - Extend probation (max 3 months additional)
 - Terminate (initiate exit clearance)
4. Update probation_end_date and employment status.

E.7.2 ID Card Request

1. Employee → My HR → ID card → Request.
2. HR verifies photo and details.
3. Generate card from HR Documents hub.
4. Record issuance date; charge replacement fee if lost (per policy).

E.7.3 Scholarship Staff (hr_portal_only)

Mining division / scholarship cohort: - Access only /my-profile scholarship sections - Cannot access sales, operations, finance modules - School fee requests via dedicated workflow - Chairman/Executive HR approves payments

E.7.4 Domestic Staff (Executive HR)

Chairman/CEO household staff: - Managed under Executive HR → Family - Separate payment workflow - Not on branch payroll runs - Confidential records

E.8 EXECUTIVE — EXTENDED PROCEDURES

E.8.1 Setting Branch Sales Targets

1. MD → Settings → Governance → Org manager targets.
2. Set monthly sales target (₹) and production metres target per branch.
3. Targets appear on Manager Dashboard pulse and Command Centre scorecard.
4. Review monthly; adjust quarterly for seasonality.

E.8.2 Credit Policy Configuration

1. Settings → Governance or org_policy_kv.
2. Set CREDIT_BRANCH_MANAGER_LIMIT_NGN.
3. Set CREDIT_MD_REQUIRED_ABOVE_NGN.
4. Set default and maximum credit terms days.
5. Communicate to all sales staff via official notice.

E.8.3 Executive Benefit Payment Workflow

draft → submitted → finance_review → md_review → approved → exported → paid

Chairman family expenses and school fees follow this chain with hr.chairman.manage.

E.9 CROSS-DEPARTMENT ESCALATION MATRIX

Issue

System down

Receipt dispute

Production delay

Refund dispute

HR grievance

Supplier quality

Coil theft suspicion

Period lock dispute

E.10 TRAINING PROGRAM OUTLINE

Week 1 — All Staff

- Sign in, password, branch selection
- Workspace navigation, official notices
- My HR self-service
- Zare introduction

Week 2 — Role Specific

- Department SOP for assigned role
- Role training guide completion
- Supervised transaction entry

Week 3 — Competency

- Checklist sign-off by supervisor
- Independent operation with spot checks

Quarterly Refresher

- Governance updates
 - New feature briefing
 - Fraud awareness
-

End of Annex E — Extended Procedures

ANNEX F: ACCOUNTING POLICIES & FINANCIAL REPORTING MANUAL

Zarewa Aluminium and Plastics Ltd — System of Operations v3.0

Authority: Finance Manager + MD approval

Reference: docs/ACCOUNTING_POLICY_AP1C.md,
docs/ACCOUNTING_POLICY_AP3_COSTING.md

F.1 ACCOUNTING POLICY OVERVIEW

Zarewa operates a three-phase accounting model implemented in the ERP:

Phase	Name	Status	Key Principle
AP1c	Production Recognition	Active (Policy V1)	Revenue at production completion
AP2	Received-Basis AP + Inventory	Active	Payables at GRN; landed cost inventory
AP3	Branch P&L + Material Costing	Progressive rollout	Standard vs actual cost per job

Phases are additive: AP3 requires AP2 requires AP1c.

F.2 POLICY V1 — REVENUE RECOGNITION (AP1c)

F.2.1 Principle

Revenue is recognised when the company has completed production of goods for the customer — not when:

- Cash is received (cash basis)
- Goods are delivered (delivery basis)
- Invoice is issued (invoice basis)

F.2.2 Journal Entries

At production job completion:

Dr Accounts Receivable (customer sub-ledger)	XXX
Cr Sales Revenue (4000)	XXX

Amount = produced metres × agreed unit price (from quotation snapshot).

At receipt clearance:

Dr Cash/Bank (1000)	XXX
Cr Accounts Receivable	XXX

At customer advance receipt (before production):

Dr Cash/Bank (1000)	XXX
Cr Customer Advances (2500)	XXX

At advance applied to quotation:

Dr Customer Advances (2500)	XXX
Cr Accounts Receivable	XXX

F.2.3 Implications for Management Reporting

Report	What It Shows
Sales (produced, MTD)	Revenue from completed jobs — may exceed cash collected
Cash & bank (cleared)	Actual money in treasury
AR as-at	Customers who owe after production recognition
Sales bridge	Reconciles quote − production − cash

MD and BM must understand: Strong production month may show high revenue before cash collection follows.

F.2.4 Refund Interaction with AP1c

Refunds reduce customer AR balance. If refund for unproduced metres after revenue recognised:

- Refund reverses portion of recognised revenue - GL impact via refund posting workflow
- Production alignment check ensures metres claim matches job records

F.3 AP2 — RECEIVED-BASIS ACCOUNTS PAYABLE

F.3.1 Principle

Supplier liabilities recognised at goods receipt (GRN), not at payment date.

F.3.2 Journal at GRN

Dr Inventory (1300)	XXX	
Cr Accounts Payable — Suppliers (2000)		XXX

At landed cost including allocated carriage inward.

F.3.3 Supplier Payment

Dr Accounts Payable (2000)	XXX
Cr Cash/Bank (1000)	XXX

F.3.4 Supplier Advances

Pre-payment before GRN:

Dr Supplier Advance (asset/sub-ledger)	XXX
Cr Cash/Bank (1000)	XXX

On GRN, advance applied:

Dr Accounts Payable (2000)	XXX
Cr Supplier Advance	XXX
Cr Cash/Bank (if balance due)	XXX

F.3.5 AP2 Diagnostics

Monthly run from Accounting desk: - Unmatched supplier advances - GRNI aging (received not paid) - Inventory GL alignment check

F.4 AP3 — MATERIAL COSTING AND BRANCH P&L

F.4.1 Purpose

Allocate true production cost per metre for branch profitability analysis.

F.4.2 Cost Components per Production Job

Component

Coil material

Accessories

Stone flatsheet

Direct labour

Overhead allocation

F.4.3 Standard vs Actual

Metric

Standard cost/m

Actual cost/m

Variance

Variance %

Report: Accounting → AP3 → Material cost report.

F.4.4 Branch P&L Structure

Line

Revenue

COGS

Gross profit

Operating expenses

Net branch contribution

Note: Inter-branch loans and HQ allocations may require manual adjustment entries.

F.5 INVENTORY VALUATION

F.5.1 Coil Inventory

- Valued at landed cost per kg
- $\text{qty_remaining} \times \text{unit cost/kg}$
- Month-end snapshot: `inventory_coil_snapshots`

F.5.2 Finished Goods

- Standard cost or last production cost per SKU
- `inventory_product_snapshots` at month-end

F.5.3 WIP

- Open production jobs: coil cost in WIP balance
- Closed at job completion to COGS

F.5.4 Closing Stock Entry

At stock register lock:

Dr Inventory — Closing (1300)	XXX
Cr COGS / Purchases (5050)	XXX

Or reverse for opening. Finance Manager determines exact mapping per period.

F.6 FIXED ASSET ACCOUNTING

F.6.1 Capitalisation Threshold

Expenditure capitalised as fixed asset when: - Category is capex (Land, Plant, Furniture, Generator) - Amount exceeds materiality threshold (per finance policy) - Future economic benefit >1 year

Auto-capitalisation on full payment for configured capex expense categories.

F.6.2 Depreciation Methods

Straight-line over useful life: - Plant & machinery: 60 months - Land & buildings: 240 months - Furniture: 84 months - Generator: 60 months

F.6.3 Disposal Accounting

Dr Accumulated Depreciation	XXX
Dr Cash (proceeds)	XXX
Dr Loss on disposal (if loss)	XXX
Cr Asset cost (150x)	XXX
Cr Gain on disposal (if gain)	XXX

F.7 CUSTOMER LEDGER ACCOUNTING

F.7.1 Ledger Entry Types

Type

RECEIPT

ADVANCE_IN

ADVANCE_APPLIED

ADVANCE_REFUND

OVERPAY_ADVANCE

OVERPAY_APPLIED

REFUND_*

F.7.2 Quotation paid_ngn Derivation

Critical rule: quotations.paid_ngn derived **only from sales receipts** — not bank reconciliation directly.

Function: syncQuotationPaidFromReceipts — run if discrepancy suspected.

F.7.3 Customer Statement

Generated from customer dashboard ledger history: - Opening balance - Production recognition (AR increase) - Receipts (AR decrease) - Refunds - Closing balance

F.8 TREASURY ACCOUNTING

F.8.1 Treasury Movement Sources

source_kind

RECEIPT

REFUND_PAYOUT

EXPENSE

SUPPLIER_PAYMENT

TRANSFER

PAYROLL

INTER_BRANCH

F.8.2 Treasury Balance

Each treasury_accounts record maintains running balance from cleared movements.

Daily check: Sum of treasury accounts = GL 1000 balance (after reconciliation).

F.9 FINANCIAL REPORTING CALENDAR

Report

Daily pack

Cash position

Weekly pack

Branch scorecard

Trial balance

Management accounts

AR aging

AP aging

Stock valuation

Payroll summary

Annual accounts

F.10 CHART OF ACCOUNTS MAINTENANCE

New GL accounts require: 1. Finance Manager proposal with account code, name, type 2.

MD approval for income statement impact 3. IT/Admin seeds in gl_accounts table 4.

Update expense category mapping if applicable 5. Update this SOP appendix

No user-created GL accounts without Finance Manager authority.

F.11 PERIOD CLOSE DETAILED CHECKLIST

Day T-5 (Before Month End)

- ☐ Notify all branches: no backdated posts after T+2 without approval
- ☐ Chase open GRNs — post all received goods
- ☐ Chase open production jobs — complete or document WIP carry
- ☐ HR confirms payroll data final if pay period

Day T (Last Business Day)

- ☐ Final receipts and payments posted
- ☐ Operations completes deliveries in transit
- ☐ Stock register physical count begins

Day T+1 to T+3

- ☐ Stock register 4-stage sign-off
- ☐ Bank reconciliation completed
- ☐ Depreciation run
- ☐ AP2/AP3 diagnostics
- ☐ Manual accrual journals

Day T+5 (Target)

- ☐ Trial balance review
 - ☐ MD reviews branch scorecard
 - ☐ **Period lock**
 - ☐ Reports pack distributed
-

F.12 TAX AND STATUTORY FILING

Obligation

PAYE remittance

Pension remittance

VAT (if applicable)

WHT certificates

Annual tax return

End of Annex F — Accounting Policies

ANNEX G: HR POLICIES, DISCIPLINE & EMPLOYEE RELATIONS MANUAL

Zarewa Aluminium and Plastics Ltd — System of Operations v3.0

Authority: GM HR + MD + Board

Reference: docs/HR/HR-POLICY-LEAVE.md, docs/HR/HR-POLICY-PAYROLL.md

G.1 EMPLOYMENT PHILOSOPHY

Zarewa is committed to fair employment practices, transparent compensation, and disciplined workforce management. The ERP HR module enforces policy through system gates — leave balances, approval chains, payroll MD sign-off, and exit clearance cannot be bypassed without administrator intervention (which is fully audited).

G.2 RECRUITMENT AND ONBOARDING

G.2.1 Manpower Planning

1. Department head submits manpower request via office memo to GM HR.
2. GM HR and MD approve headcount and budget.
3. HR Admin creates job posting at /hr/talent → Recruit.
4. Public listing at /api/public/careers/* (unauthenticated).

G.2.2 Selection Process

Stage

Application receipt

Shortlisting

Interview

Offer

Acceptance

Onboarding

G.2.3 Onboarding Checklist

- ☐ Staff profile created with all required fields (\$13.2 SOP-07)
- ☐ ERP login created (Settings → Team & access)
- ☐ Role and branch assigned
- ☐ selfServiceEligible = true
- ☐ Bank details entered and verified
- ☐ NOK recorded
- ☐ Probation end date set (default +6 months)
- ☐ Policy acknowledgements sent
- ☐ ID card request initiated

- ☐ Asset custody assigned (laptop, phone, uniform if applicable)
 - ☐ Department SOP training scheduled
 - ☐ Line manager introduced on Team HR
-

G.3 COMPENSATION STRUCTURE

G.3.1 Salary Levels

Level

L1

L2

L3

L4

L5

L6

L7

G.3.2 Salary Matrix

Maintained at /hr/payroll → Salary matrix: - Rows: levels L1–L7 - Columns: steps 1–5 (annual increment progression) - Changes require salary_structure.approve (MD)

G.3.3 Allowances

Allowance

Transport

Housing

Meal subsidy

Shift premium

Acting allowance

G.3.4 Special Increments

Outside annual increment cycle: 1. Department head submits justification memo. 2. GM HR reviews. 3. MD approves via special_increment.approve. 4. HR Admin updates salary matrix effective date. 5. Reflected in next payroll run.

G.4 LEAVE POLICY (DETAILED)

G.4.1 Annual Leave Accrual

- Accrual: monthly via hr_leave_accrual_ledger
- Junior band: $14 \text{ days} \div 12 = 1.17 \text{ days/month}$
- Senior band: $21 \text{ days} \div 12 = 1.75 \text{ days/month}$

- Maximum carry-forward: 5 days (configurable in org policy)
- Leave encashment: on separation only; GM HR approval

G.4.2 Sick Leave

- No fixed balance
- 1–2 days: self-certification
- 3+ days: medical certificate required
- Extended illness: HR welfare visit; may trigger disability review
- Unpaid if statutory entitlement exhausted

G.4.3 Maternity Leave

- Days: maternityLeaveDays in org_policy_kv
- Paid per company policy and statutory minimum
- HR plans cover during absence
- Return-to-work interview at 2 weeks before resumption

G.4.4 Compassionate Leave

- Bereavement: spouse/parent/child — 3 days paid
- Extended family — 1 day paid
- GM HR discretion for additional unpaid days

G.4.5 Leave Planning

- Peak season (rainy season roofing demand): BM may restrict leave to 20% of team simultaneously
 - Blackout dates published via official notice
 - Leave calendar visible at /hr/time-absence → Calendar
-

G.5 STAFF LOAN POLICY

G.5.1 Eligibility

Criterion

Minimum service

Discipline record

Existing obligation

Maximum amount

Maximum term

G.5.2 Approval Chain

Employee → HR review → BM endorse → GM HR final → (MD if above ₦1,000,000 exceptional)

G.5.3 Interest

Per company policy (configurable): - Emergency loans: 0% interest - General loans: 5% flat or as per board resolution - Recorded in loan agreement PDF (POST /api/hr/loan-requests/:id/agreement-letter)

G.5.4 Default

3 consecutive missed payroll deductions: 1. HR issues demand letter 2. Discipline case if negligence 3. MD may authorise full balance acceleration 4. Write-off requires MD + Board if unrecoverable

G.6 STAFF PURCHASE CREDIT (HR PERSPECTIVE)

Linked to SOP-01 §8 — HR verifies: - 12 months service - No active discipline case - OBL capacity check (max ₦5M outstanding) - BM commercial endorsement - Payroll deduction schedule agreed before disbursement

G.7 ATTENDANCE AND TIME MANAGEMENT

G.7.1 Working Hours

Category

Head office

Factory

Sales

G.7.2 Attendance Recording

- Daily roll: Team HR or HR Admin marks at /team-hr/time-absence or /hr/time-absence
- Upload: bulk biometric export via hr_attendance_uploads
- Late arrival: flagged after grace period (15 minutes)
- Absent without leave: triggers discipline workflow

G.7.3 Payroll Impact

Attendance Event

Unauthorised absence

Late (3+ in month)

Approved leave

Unpaid leave

G.8 DISCIPLINE POLICY (DETAILED)

G.8.1 Principles

- Progressive discipline except gross misconduct
- Right to respond in writing at written warning stage
- HR present at all formal disciplinary meetings
- All cases recorded in ERP — no informal “off record” warnings for serious matters

G.8.2 Gross Misconduct (Summary Dismissal)

Offence

Theft/fraud

Violence

Gross insubordination

File tampering

Serious negligence

Conflict of interest

Intoxication at work

Process: Suspension pending investigation → hearing → MD decision → exit clearance

G.8.3 Investigation Procedure

1. GM HR appoints investigator (not the direct supervisor if conflicted).
2. Case status: under_investigation.
3. Witness statements recorded in hr_discipline_cases evidence.
4. Employee given opportunity to respond.
5. Investigation report to MD.
6. Decision: no action, warning, suspension, dismissal.

G.8.4 Appeal

Employee may appeal within 7 days of action issued: - Appeal to MD (if BM decided) or Board (if MD decided) - Status: appealed - Outcome: uphold, reduce, or overturn

G.9 GRIEVANCE PROCEDURE

1. Employee submits grievance via My HR → Feedback/Grievance.
2. HR acknowledges within 48 hours.
3. Informal resolution attempted with line manager.
4. If unresolved: formal hearing with HR + neutral manager.
5. GM HR decision within 10 working days.
6. Escalation to MD if policy matter.
7. All records confidential — workspaceConfidentialAccess.

G.10 PERFORMANCE MANAGEMENT

G.10.1 Appraisal Cycle

- Annual appraisal for all permanent staff
- Probation review at 6 months
- Recorded in hr_performance_reviews

G.10.2 Appraisal Process

Step

Self-assessment

Manager rating

Calibration

Final rating

Development plan

G.10.3 Performance-Linked Actions

Rating

Outstanding

Meets expectations

Needs improvement

Unsatisfactory

G.11 TRANSFER AND SECONDMENT

G.11.1 Policy (from transferTenurePolicy.js)

Type

Inter-branch

Internal rotation

BM transfer

G.11.2 Transfer Procedure

1. Employee or manager initiates transfer request.
 2. Receiving branch BM agrees to receive.
 3. GM HR reviews manpower impact.
 4. MD approves inter-branch.
 5. HR updates branch assignment and ERP workspace branch.
 6. Asset custody transferred.
 7. Salary matrix unchanged unless promotion attached.
-

G.12 SEPARATION AND EXIT (DETAILED)

G.12.1 Types of Separation

Type

Resignation

Termination (performance)

Dismissal (gross misconduct)

Redundancy

End of contract

Retirement

G.12.2 Exit Clearance Detail

Stage 1 — Initiate (HR Admin) - Reason code, last working day, notice waiver if applicable

Stage 2 — HR Clearance - ID card returned - Outstanding loans documented - Documents collected (handbook, keys) - Exit interview scheduled

Stage 3 — Finance Clearance - OBL balance settled or deduction agreement - Final payroll calculated (pro-rata + leave encashment) - Company property fines if applicable

Stage 4 — Admin Clearance - IT access revocation request to Admin - Asset custody returned (laptop, phone, uniform) - Handover document filed

Stage 5 — Final Clearance (GM HR / MD) - All stages green - Separation letter generated - ERP account deactivated - Certificate of service issued

G.12.3 Final Payroll

Includes: - Pro-rata salary to last day - Accrued leave encashment (if eligible) - Less: outstanding loan balance - Less: asset recovery - PAYE and pension on final amounts

G.13 EXECUTIVE AND BOARD MEMBER COMPENSATION

G.13.1 Director Emoluments

- director_emolument compensation type
- Board-approved packages
- Processed via executive payroll track
- MD and Chairman view via Executive HR → Compensation

G.13.2 Chairman Family Benefits

- School fees: hr_chairman_school_fees
- Family expenses: hr_chairman_expenses

- Managed by Chairman/CEO with hr.chairman.manage
 - Separate from branch payroll
 - Finance review + MD approval on payment workflow
-

G.14 HR REPORTING

Report

Headcount by branch

Leave balances

Loan outstanding

Discipline summary

Probation tracker

Document expiry

Attendance register

Payroll summary

Branch HR contribution

G.15 HR DATA PROTECTION

- Access to staff profiles: HR permissions only
 - Bank details encrypted (hrBankCrypto.js)
 - Salary data: payroll.view_sensitive for executive roles
 - Discipline cases: confidential work items
 - Exit records: retained 7 years minimum
 - Right of access: employee may request copy of HR file via HR Admin
-

End of Annex G — HR Policies Manual

ANNEX H: INVENTORY, COIL SCIENCE & PRODUCTION STANDARDS

Zarewa Aluminium and Plastics Ltd — System of Operations v3.0

Audience: Operations, production, procurement, finance

H.1 COIL MATERIAL SCIENCE

H.1.1 Material Types

Type

Aluzinc (PPGI)
Aluminium
Stone-coated substrate

H.1.2 Gauge Selection Guide

Gauge (mm)

0.18
0.22
0.30
0.45
0.50

Sales must quote gauge per customer structural requirement — never down-gauge without written customer acknowledgement.

H.1.3 Standard Conversion Factors (kg/m)

Maintained in Procurement → Conversion tab. Example reference values (verify live system):

Gauge (mm)

0.18
0.22
0.30
0.45
0.50

Theoretical metres = kg consumed ÷ kg/m factor

H.2 COIL RECEIPT INSPECTION

H.2.1 On Arrival Checklist

- ☐ PO reference matches delivery note
- ☐ Coil numbers match supplier manifest
- ☐ Weighbridge weight within PO tolerance (typically $\pm 2\%$)
- ☐ Gauge verified with micrometer (sample)
- ☐ Colour matches PO specification (visual)
- ☐ No edge damage, rust, or oil contamination
- ☐ Mill test certificate collected

H.2.2 Rejection Criteria

Reject coil (do not GRN) if: - Weight short > agreed tolerance - Wrong gauge or colour - Visible lamination defect - Water damage

Notify Procurement and MD; photograph damage; supplier claim initiated.

H.3 PRODUCTION STANDARDS

H.3.1 Corrugation Profiles

Profile

Long-span

Step-tile

Custom

H.3.2 Quality Standards

Parameter

Profile depth

Cover width

Colour uniformity

Edge condition

Metre count

H.3.3 Stone-Coated Production

1. Substrate coil → corrugation → stone application
 2. Sheets counted; $m^2 = \text{sheets} \times \text{area per sheet}$
 3. stoneFlatSheetSheetsToM2 conversion in system
 4. Quality: stone adhesion test per batch (QCO)
-

H.4 OFFCUT MANAGEMENT

H.4.1 Normal Offcut

Production waste within 3% of theoretical — no material incident required.

H.4.2 Abnormal Offcut

Waste >3% or identifiable defect: 1. Log material incident (production_offcut type) 2. BM approves 3. Metres added to offcut pool 4. Available for smaller orders or internal use

H.4.3 Offcut Issuance

When production uses offcut pool metres: 1. Select incident on job completion 2. System decrements meters_available on incident 3. Completion record shows “supplied from offcut” 4. May affect costing (AP3) — lower material cost

H.5 STOCK REGISTER PROCEDURE (DETAILED)

H.5.1 Physical Count Scope

Asset Class

Coil lots

Finished goods

Accessories

Stone stock

WIP

H.5.2 Variance Investigation

Variance

Coil kg

FG metres

Accessories

H.5.3 BM Adjustment Window

Between store_confirmed and bm_approved: - BM may post adjustment entries with documented reason - After bm_approved: adjustments require MD and new count cycle

H.6 DELIVERY STANDARDS

H.6.1 Loading

- Secure strapping per safety standard
- Gauge/colour labels visible on bundles
- Delivery note matches system DLV lines
- Vehicle capacity not exceeded

H.6.2 Proof of Delivery

Required fields on POD: - Customer name and signature - Date and time - Metres delivered per line - Condition notes (damage at delivery) - Driver name

POD scanned and attached to delivery record in system.

End of Annex H — Inventory & Production Standards